An exposition of a computer-assisted proof of the bound $\Lambda \le 0.1787854$ for the de Bruijn–Newman constant

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Abstract

We give a self-contained mathematical exposition of a claimed computer-assisted proof, due to J. Gomila, of the bound

$$\Lambda \le \frac{893927}{5000000} = 0.1787854$$

for the de Bruijn–Newman constant Λ, improving the bound $\Lambda \le 0.22$ of the Polymath 15 project and the current record $\Lambda \le 0.2$ of Platt–Trudgian. The argument is intended to be unconditional: it instantiates the criterion of Polymath 15 [Theorem 1.2, arXiv:1904.12438v2] at the exact parameters $X = 6000000185827$, $t_0 = 129/800$, $y_0^2 = 87677/2500000$, supplying its three hypotheses by, respectively, the Platt–Trudgian finite verification of the Riemann hypothesis, a hybrid finite-scan/analytic-tail nonvanishing theorem for H_{t_0}, and an interval-arithmetic barrier certificate. The proof has not been peer reviewed. In the course of the review recorded here, every analytic lemma has been given a complete verified proof; all statements quoted from the published literature have been checked against the arXiv sources of the cited papers; and each of the five computer-assisted components has been independently reproduced, by programs reviewed line by line against the corresponding statements, running on an independently built toolchain and reading no stored data. The verification status is summarized in Section 8; a final overall assessment is deferred to the conclusion of the review.

Contents

*Referee's exposition of an unpublished proof by J. Gomila (private repository `dbn-lambda-01787854-candidate-audit`, "sealed referee snapshot," 2026). This document describes the argument for the purpose of independent review. At the present stage of the review every analytic lemma carries a complete verified proof and every computer-assisted component has been independently reproduced (see Section 8); a final overall certification of correctness awaits the completion of the review.

1 Introduction

1.1 The de Bruijn–Newman constant

Let Φ denote the super-exponentially decaying function

$$\Phi(u) = \sum_{n=1}^{\infty} \big(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}\big) \exp\!\big(-\pi n^2 e^{4u}\big),$$

and for $t \in \mathbb{R}$ define the entire functions

$$H_t(z) = \int_0^{\infty} e^{tu^2} \Phi(u) \cos(zu)\, du. \tag{1}$$

The function H_0 is a normalization of the Riemann ξ-function: with $\xi(s) = \frac{s(s-1)}{2}\pi^{-s/2}\Gamma(s/2)\zeta(s)$ one has $H_0(x) = \frac{1}{8}\,\xi\big(\frac{1}{2} + \frac{ix}{2}\big)$, so that the Riemann hypothesis (RH) is the statement that all zeros of H_0 are real. De Bruijn [1] showed that if H_t has only real zeros for some t, the same holds for all larger t; Newman [5] showed that for some t the function H_t has a non-real zero. Hence there is a constant $\Lambda \in \mathbb{R}$, the *de Bruijn–Newman constant*, such that H_t has only real zeros if and only if $t \geq \Lambda$. RH is equivalent to $\Lambda \leq 0$.

Newman conjectured $\Lambda \geq 0$, which was proved by Rodgers and Tao [8]. On the other side, a long sequence of numerical upper bounds (de Bruijn's $\Lambda \leq 1/2$, then [2, 3] and others) led to the bound

$$\Lambda \leq 0.22 \tag{2}$$

of the Polymath 15 project [7]. Subsequently Platt and Trudgian observed [6, Corollary 2] that their verification of RH to height $3 \cdot 10^{12}$ (Section 3 below), combined with the second row of the parameter table in [7, Table 1, p. 65] (which requires verified height $H > 2.51 \cdot 10^{12}$), yields the current record

$$\Lambda \leq 0.2. \tag{3}$$

Thus at present $0 \leq \Lambda \leq 0.2$. Platt and Trudgian further note that the next entry of the same table would give $\Lambda < 0.19$, but only conditionally on a verified height slightly above 10^{13}, beyond their theorem; the bound (5) below therefore goes past both the published record (3) and that conditional entry, while consuming only the established $3 \cdot 10^{12}$ verification height.

1.2 Main result

This paper describes a claimed computer-assisted proof, due to J. Gomila, of the following improvement of (3).

Theorem 1.1 (Theorem; **unreviewed**).

$$\Lambda \leq \frac{893927}{5000000} = 0.1787854.$$

The proposed proof instantiates the barrier criterion of [7, Theorem 1.2] at the exact parameters

$$X = 6000000185827, \qquad t_0 = \frac{129}{800} = 0.16125, \qquad y_0^2 = \frac{87677}{2500000} = 0.0350708, \tag{4}$$

with $y_0 > 0$, so that

$$t_0 + \frac{y_0^2}{2} = \frac{893927}{5000000} = 0.1787854. \tag{5}$$

One checks the elementary inequalities $0 < t_0 < \frac{1}{2}$, $0 < y_0^2 < 1 - 2t_0 = \frac{271}{400}$, and $y_0^2 + 2t_0 = \frac{893927}{2500000} < 1$, which place the parameters in the domain of the criterion.

1.3 Status, and how gaps are marked

The claimed proof combines:

1. two published theorem inputs — the Polymath 15 criterion and effective approximation [7], and the Platt–Trudgian finite verification of RH [6];
2. a collection of new lemmas (reproduced with proofs below), which bridge the published estimates to the quantities computed by the software; and
3. a large body of rigorous-interval-arithmetic certificates (FLINT/Arb in C, plus corroborating Python interval implementations), totalling 3,149,013 certified finite "window" rows, a standalone tail computation, and an 883-prism barrier certificate.

The statements quoted from the published inputs in item (1) — Theorems 1.2 and 1.3 of [7] together with the displays (14), (20)–(24), (71)–(72), Proposition 6.6 and Lemma 8.4 of that paper, and Theorem 1 of [6] — have been checked verbatim against the arXiv source files of the cited papers (1904.12438v2 and 2004.09765v1, to which all equation and theorem numbers below refer). The one point where the current proof's usage deliberately departs from a cited display, namely equation (24) of [7], is discussed in Remark 2.2. The lemmas of item (2) are given complete proofs below, all verified in the course of this review. Every computation in item (3) — the main finite computation (Proposition 4.3), the Dini cell computation (Proposition 4.9), the finite-region error bound (Proposition 4.10), the tail cutoff computation (Proposition 5.10), the barrier-box error bound (Proposition 6.2), the tail-exponent inequality (Proposition 6.5), and the barrier prism decomposition (Proposition 6.12) — has been reproduced on an independent toolchain, by a standalone program (supplementary directory `code/`) that reads no stored data and was reviewed line by line against the corresponding statement; the proofs are given with the respective propositions. No unverified steps remain flagged; the earlier framed *gap / review points* of this document have all been resolved and removed. The verification is summarized in Section 8.

2 The Polymath criterion and effective approximation

2.1 The barrier criterion

The logical skeleton of the proof is the following criterion.

Theorem 2.1 (Polymath [7, Theorem 1.2])**.** *Let $X > 0$, $0 < t_0 < \frac{1}{2}$ and $0 < y_0^2 < 1 - 2t_0$. Suppose that:*

(i) ***(Verified height.)*** *There are no zeros $\zeta(\sigma + iT) = 0$ with*

$$\frac{1+y_0}{2} \le \sigma \le 1, \qquad 0 \le T \le \frac{X}{2}.$$

(ii) ***(Final-time right region.)*** *There are no zeros $H_{t_0}(x + iy) = 0$ with*

$$x \ge X + \sqrt{1 - y_0^2}, \qquad y_0 \le y \le \sqrt{1 - 2t_0}.$$

(iii) ***(Intermediate-time barrier.)*** *There are no zeros $H_t(x + iy) = 0$ with*

$$X \le x \le X + \sqrt{1 - y_0^2}, \qquad \sqrt{y_0^2 + 2(t_0 - t)} \le y \le \sqrt{1 - 2t}, \qquad 0 \le t \le t_0.$$

Then

$$\Lambda \le t_0 + \frac{y_0^2}{2}.$$

Sections 3, 4–5, and 6 below describe the proof's suppliers for hypotheses (i), (ii), and (iii) respectively, at the parameters (4).

For hypothesis (i) one uses the normalization connecting H_0 to ζ: a zero $H_0(x + iy) = 0$ gives $\xi\left(\frac{1-y+ix}{2}\right) = 0$; the functional equation $\xi(s) = \xi(1 - s)$ followed by complex conjugation produces the representative zero $\xi\left(\frac{1+y+ix}{2}\right) = 0$ with $\sigma = \frac{1+y}{2} \ge \frac{1+y_0}{2}$ and ordinate $T = \frac{x}{2}$, which is the region excluded in (i). (The proof verifies the affine algebra of this "sign map" by an exact-rational script; the functional equation and the reality symmetry $\overline{\xi(\bar{s})} = \xi(s)$ are the classical inputs.)

2.2 The effective approximation

For $t > 0$ write, following [7],

$$g_t(z) = \frac{H_t(z)}{B_t(z)}, \tag{6}$$

where B_t is an explicit normalizing factor: in the notation of [7], $B_t(x+iy) = M_t\big(\frac{1+y-ix}{2}\big)$, where $M_t(s) = \exp\big(\frac{t}{4}\alpha(s)^2\big)M_0(s)$ and M_0 is defined in [7, (6)] as the exponential of an explicit holomorphic branch of its logarithm on $\mathbb{C}\setminus(-\infty,1]$ (see [7, (7)]). In particular M_0, and hence M_t, takes values in $\mathbb{C}\setminus\{0\}$ on its domain; since the argument $\frac{1+y-ix}{2}$ is non-real whenever $x \neq 0$, it avoids the branch cut, and therefore $B_t(x+iy) \neq 0$ for all $t \geq 0$, $x > 0$ and $y > 0$, as noted explicitly in [7, §1, after (11)]. This nonvanishing is used below both in the reduction of zero-freeness of H_t to lower bounds on $|f_t|$ and in the argument-principle step of the barrier certificate (Section 6). Theorem 1.3 of [7] provides, in the region

$$x \geq 200, \qquad 0 \leq y \leq 1, \qquad 0 < t \leq \tfrac{1}{2}, \tag{7}$$

an approximation

$$g_t(x+iy) = f_t(x+iy) + O_{\leq}\big(e_A + e_B + e_{C,0}\big), \tag{8}$$

where $O_{\leq}(E)$ denotes a quantity of absolute value at most E, the main term is the finite sum (equation (14) of [7])

$$f_t(x+iy) = \sum_{n=1}^{N} b_t(n)\, n^{-s_*} + \gamma \sum_{n=1}^{N} n^y\, b_t(n)\, n^{-\overline{s_*}-\kappa}, \qquad b_t(n) = \exp\Big(\frac{t}{4}\log^2 n\Big), \tag{9}$$

(we write $b_t(n)$ for the quantity denoted b_n^t in [7]; all other symbols in (9) agree with the notation there, including the placement of κ alongside the conjugated exponent $\overline{s_*}$ in the second sum) with:

- the window index $N = \big\lfloor \sqrt{\frac{x}{4\pi} + \frac{t}{16}} \big\rfloor$;
- explicitly defined quantities $\gamma = \gamma(x,y,t)$, $s_* = s_*(x,y,t)$, $\kappa = \kappa(x,y,t)$ satisfying (by equations (20)–(22) of [7]) the bounds

$$|\gamma| \;\leq\; e^{0.02y}\Big(\frac{x}{4\pi}\Big)^{-y/2}, \tag{10}$$

$$\operatorname{Re} s_* \;\geq\; \frac{1+y}{2} + \frac{t}{4}\log\frac{x}{4\pi} - \frac{t}{2x^2}\Big(1 - 3y + \frac{4y(1+y)}{x^2}\Big)_+, \tag{11}$$

$$|\kappa| \;\leq\; \frac{ty}{2(x-6)}; \tag{12}$$

- and error terms $e_A, e_B, e_{C,0}$ with explicit majorants (equations (71)–(72) and (23)–(24) of [7]).

Since $B_t \neq 0$, nonvanishing of H_t on a region follows from the strict inequality $|f_t| > e_A + e_B + e_{C,0}$ there; this implication is itself stated in [7] as Corollary 1.4 ("Criterion for non-vanishing").

2.3 The conservative error constant 10.50

The current proof deliberately does *not* use the constant 10.44 displayed in equation (24) of [7] for the majorization of $e_{C,0}$. Instead it derives a slightly weaker bound directly from Proposition 6.6(vi) of [7], which contains the expression

$$\exp\left(\frac{3\left|\log\frac{x}{4\pi}+\frac{i\pi}{2}\right|+3.58}{x-8.52}\right)\left(1+\frac{1.24\,(3^{y}+3^{-y})}{N-0.125}+\frac{6.92}{x-12}\right).$$

On the theorem domain $x \geq 200$ one has $x-8.52 > x-12 > 0$; applying $1+u \leq e^{u}$ to the second factor and enlarging the denominator $x-8.52$ to $x-12$ yields the *conservative corollary*

$$e_{C,0} \leq \left(\frac{x}{4\pi}\right)^{-(1+y)/4}\exp\left(-\frac{t}{16}\log^{2}\frac{x}{4\pi}+\frac{1.24\,(3^{y}+3^{-y})}{N-0.125}+\frac{3\left|\log\frac{x}{4\pi}+\frac{i\pi}{2}\right|+10.50}{x-12}\right), \tag{13}$$

with $3.58+6.92=\frac{21}{2}=10.50$ exactly. Every numerical error computation below (finite range, tail, and barrier) uses (13), and uses the denominator $x-6.66$, which appears in equation (23) and in Proposition 6.6(iv),(v) of [7], for e_A+e_B.

Remark 2.2. Equation (24) of [7] displays the bound (13) with the constant 10.44 in place of 10.50. The sentence following Proposition 6.6 in [7] states that (24) is obtained from part (vi) of the proposition by "the inequality $1+u \leq \exp(u)$" followed by the bound $\frac{1}{x-8.52} \leq \frac{1}{x-12}$; carrying out this recipe yields the constant $3.58+6.92=10.50$, and we are not aware of an argument in [7] producing the displayed 10.44. Accordingly, the current proof does not rely on (24): all error computations use only (13), which follows rigorously from Proposition 6.6(vi) by the two elementary steps just described, uniformly on the domain (7). (The weaker constant costs a factor $e^{0.06/(x-12)}$, which is utterly negligible at $x \approx 6\times 10^{12}$.)

3 Hypothesis (i): the verified height

Platt and Trudgian [6, Theorem 1] proved that every nontrivial zero of ζ with ordinate in $(0, T_{\mathrm{PT}}]$,

$$T_{\mathrm{PT}} = 3000175332800,$$

lies on the critical line. (Their statement reads: "The Riemann hypothesis is true up to height 3 000 175 332 800. That is, the lowest 12 363 153 437 138 non-trivial zeroes ρ have $\Re\rho = 1/2$." The method — counting sign changes of the completed zeta function on the critical line and confirming by Turing's method that all zeros are accounted for — yields the off-line exclusion form used here.) Hypothesis (i) of Theorem 2.1 requires the absence of zeros with $\sigma \geq \frac{1+y_0}{2} > \frac{1}{2}$ and $0 \leq T \leq X/2$. Since

$$\frac{X}{2} = 3000000092913.5, \qquad T_{\mathrm{PT}} - \frac{X}{2} = \frac{350479773}{2} > 0,$$

the verified region covers all ordinates $0 < T \leq X/2$ with margin $\approx 1.75\times 10^{8}$. On the segment $T=0$, $\frac{1}{2} < \sigma < 1$, one argues directly: the Dirichlet eta function satisfies $\eta(\sigma) = \sum_{n\geq 1}(-1)^{n-1}n^{-\sigma} > 0$ for $0<\sigma<1$ (group consecutive terms), while $\eta(\sigma) = (1-2^{1-\sigma})\zeta(\sigma)$ with $1-2^{1-\sigma} < 0$; hence $\zeta(\sigma) < 0 \neq 0$ there. At $\sigma = 1$, ζ has a pole. This supplies hypothesis (i).

As with any use of a computationally established theorem, this step inherits the correctness of the Platt–Trudgian computation itself (a rigorous interval-arithmetic computation, independent of anything in the present paper); we accept it here as a published-theorem input.

4 Hypothesis (ii), finite range: the window scan

Hypothesis (ii) concerns nonvanishing of H_{t_0} on

$$x \geq x_* := X + \sqrt{1 - y_0^2}, \qquad y_0 \leq y \leq \sqrt{1 - 2t_0}. \tag{14}$$

Gomila's proof splits this region at the abscissa corresponding to the window index $N_* = 3840000$: the finite range $690988 \leq N \leq 3840000$ is handled in this section by 3,149,013 individually certified interval computations, and the infinite range $N \geq 3840000$ in Section 5 by a single analytic contraction argument. The two ranges overlap on a full window, eliminating endpoint risk.

4.1 Windows and the freeze lemma

At $t = t_0$ define

$$x_N = 4\pi\Big(N^2 - \frac{t_0}{16}\Big), \qquad W_N = [x_N, x_{N+1}),$$

so that $x \in W_N$ iff the window index of x equals N. Exact rational Machin-type bounds for π and exact integer square-root brackets (independently confirmed by a 400-bit interval computation) give

$$x_{690988} < x_* < x_{690989}, \tag{15}$$

with margins exceeding 5.37×10^6 and 1.19×10^7 respectively; thus x_* lies strictly inside W_{690988}, and the half-open windows tile $[x_*, x_{3840001})$ with no gaps or overlaps.

The certified computations evaluate, for each N, the majorants of (10)–(12) *frozen at the left edge* $x = x_N$ of the window. The following elementary lemma justifies this.

Lemma 4.1 (Window freeze)**.** *For $y_0 \leq y \leq \sqrt{1 - 2t_0}$ define*

$$\begin{aligned}
G(x, y) &= e^{y/50}\Big(\frac{x}{4\pi}\Big)^{-y/2},\\
K(x, y) &= \frac{t_0 y}{2(x - 6)},\\
\Sigma(x, y) &= \frac{1 + y}{2} + \frac{t_0}{4}\log\frac{x}{4\pi} - \frac{t_0}{2x^2}\Big(1 - 3y + \frac{4y(1 + y)}{x^2}\Big)_+.
\end{aligned}$$

Then for every N in the finite range, every $x \in W_N$ and every y in the stated range,

$$G(x, y) \leq G(x_N, y), \qquad K(x, y) \leq K(x_N, y), \qquad \Sigma(x, y) \geq \Sigma(x_N, y).$$

Proof. $\partial_x G = -\frac{y}{2x}G < 0$ and $\partial_x K = -\frac{t_0 y}{2(x-6)^2} < 0$ give the first two inequalities. For the third, write $h(x, y) = 1 - 3y + A(y)/x^2$ with $A(y) = 4y(1 + y) \geq 0$; then $h_x = -2A/x^3 \leq 0$, so h_+ is nonincreasing in x, and since x^{-2} is positive and decreasing, the product $x^{-2}h_+$ is nonincreasing. The logarithmic term is strictly increasing, so Σ is strictly increasing in x, including across the kink $h = 0$ (both one-sided derivatives there are positive). □

Note that $e^{y/50} = e^{0.02y}$, matching (10); G, K, Σ frozen at x_N are precisely the quantities g_N (bound for $|\gamma|$), k_N (bound for $|\kappa|$) and σ_N (lower bound for $\operatorname{Re} s_*$) used in the definition of the certified quantities of Section 4.3.

4.2 The mollified lower bound for $|f_{t_0}|$

Fix a window N, a height $y > 0$, and a finite set of auxiliary primes $\mathcal{P}$ with $D = \prod_{p\in\mathcal{P}} p$. Let g_N, k_N, σ_N be as above, and introduce the real *Euler mollifier*

$$E(s_*) = \prod_{p\in\mathcal{P}} \bigl(1 - b_{t_0}(p)\, p^{-s_*}\bigr) = \sum_{d|D} \lambda_d\, d^{-s_*}, \qquad \lambda_d = (-1)^{\omega(d)} \prod_{p|d} b_{t_0}(p) \in \mathbb{R}.$$

Define the convolved coefficients (for $1 \le n \le DN$)

$$B_{N,n} = \sum_{\substack{d|D,\ d|n \\ n/d\le N}} \lambda_d\, b_{t_0}(n/d), \qquad A_{N,n} = \sum_{\substack{d|D,\ d|n \\ n/d\le N}} \lambda_d\, b_{t_0}(n/d)\,(n/d)^y,$$

so $B_{N,1} = A_{N,1} = 1$, and set

$$M_N = \sum_{d|D} |\lambda_d|\, d^{-\sigma_N}, \qquad C_N = \sum_{m=2}^{N} b_{t_0}(m)\bigl(m^{k_N} - 1\bigr) m^{y-\sigma_N},$$

$$L_N = \frac{1 - g_N - \sum_{n=2}^{DN} \bigl(|B_{N,n}| + g_N |A_{N,n}|\bigr) n^{-\sigma_N}}{M_N} \; - \; g_N C_N. \tag{16}$$

Lemma 4.2 (Mollified lower bound)**.** *If $L_N > 0$, then $|f_{t_0}(x+iy)| \ge L_N$ for all $x \in W_N$.*

Proof. Write (9) as $f_t = A + \gamma C_\kappa$ with $A = \sum_{m\le N} b_t(m) m^{-s_*}$ and $C_\kappa = \sum_{m\le N} m^y b_t(m) m^{-\overline{s_*}-\kappa}$. Exact Dirichlet convolution gives the two identities

$$E(s_*)\, A = \sum_{n=1}^{DN} B_{N,n}\, n^{-s_*}, \qquad \overline{E(s_*)}\, C_0 = \sum_{n=1}^{DN} A_{N,n}\, n^{-\overline{s_*}}, \tag{17}$$

where the second identity uses that the λ_d are *real* (so that $\overline{E(s_*)} = \sum_d \lambda_d d^{-\overline{s_*}}$). By the triangle inequality and $\operatorname{Re} s_* \ge \sigma_N$,

$$|EA| \ge 1 - \sum_{n=2}^{DN} |B_{N,n}| n^{-\sigma_N}, \qquad |\overline{E} C_0| \le 1 + \sum_{n=2}^{DN} |A_{N,n}| n^{-\sigma_N}.$$

For complex κ with $|\kappa| \le k_N$,

$$|m^{-\kappa} - 1| = \bigl|e^{-\kappa \log m} - 1\bigr| \le e^{|\kappa| \log m} - 1 \le m^{k_N} - 1, \tag{18}$$

whence $|\overline{E}\,(C_\kappa - C_0)| \le |E|\, C_N$. Combining, and using $|\gamma| \le g_N$,

$$|E|\,|f_t| \;\ge\; Q_N - g_N |E| C_N, \qquad Q_N := 1 - g_N - \sum_{n=2}^{DN} \bigl(|B_{N,n}| + g_N |A_{N,n}|\bigr) n^{-\sigma_N}.$$

If $L_N > 0$ then $Q_N = M_N(L_N + g_N C_N) > 0$; the displayed inequality then forces $E \ne 0$ (otherwise $0 \ge Q_N > 0$). Dividing by $|E|$ and using $|E| \le M_N$ together with $Q_N > 0$,

$$|f_t| \;\ge\; \frac{Q_N}{|E|} - g_N C_N \;\ge\; \frac{Q_N}{M_N} - g_N C_N = L_N. \qquad \square$$

The point of Lemma 4.2 is that any certified per-window lower bound for L_N is *already* in the same normalized units as the additive error in (8); no further Euler-factor conversion intervenes.

4.3 The certified lower bounds

For each integer N with $690988 \leq N \leq 3840000$, define a set of auxiliary primes

$$\mathcal{P}(N) = \begin{cases} \{2,3,5,7,11\}, & 690988 \leq N \leq 728999, \\ \{2,3,5,7\}, & 729000 \leq N \leq 818999, \\ \{2,3,5\}, & 819000 \leq N \leq 1027999, \\ \{2,3\}, & 1028000 \leq N \leq 3840000, \end{cases} \tag{19}$$

and let L_N denote the quantity (16) formed with the prime set $\mathcal{P}(N)$, at $t = t_0$ and $y = y_0$, with the constants $g_N = G(x_N, y_0)$, $k_N = K(x_N, y_0)$, $\sigma_N = \Sigma(x_N, y_0)$ frozen at the window's left edge as in Lemma 4.1. (The number of terms in L_N grows with $|\mathcal{P}(N)|$; the choice (19) uses the larger, more effective mollifiers only on the ranges of N where they are needed.)

Proposition 4.3 (Certified lower bounds for L_N)**.** *For every integer* $690988 \leq N \leq 3840000$,

$$L_N \;\geq\; T_{\min} \;=\; 791366 \times 10^{-12} \;>\; 0.$$

More precisely, the minimum of L_N over each of the four ranges in (19) *is bounded below as listed in Table 1; the global bound $T_{\min}$ is attained at* $N = 690988$.

$\mathcal{P}(N)$	range of N	certified lower bound for $\min L_N$
$\{2,3,5,7,11\}$	$690988 \leq N \leq 728999$	0.000000791366
$\{2,3,5,7\}$	$729000 \leq N \leq 818999$	0.000315112481
$\{2,3,5\}$	$819000 \leq N \leq 1027999$	0.000305788816
$\{2,3\}$	$1028000 \leq N \leq 3840000$	0.000309285478

Table 1: The auxiliary prime sets (19) and, for each of the four ranges of N, the certified lower bound for $\min_N L_N$ established by the verification run described in the proof of Proposition 4.3.

Proof. The proof is computer-assisted; the verification program `prop43_proof.c`, included with this manuscript as supplementary material (directory `code/prop43`), computes for each N an outward-rounded interval enclosure of a quantity provably at most L_N, and prints the floor of 10^{12} times the lower endpoint of that enclosure. Each printed value is therefore a certified lower bound for L_N. The program is derived from Gomila's source (as documented in its header) and was reviewed line by line as part of this audit. It has two evaluation modes: a direct term-by-term interval evaluation of (16), and a production sweep that majorizes the sum in (16) by activation-frozen weights (conservative because the factor g decreases in N) and evaluates the result through certified Taylor moments with an explicit Lagrange remainder. The review verified that every step of both modes is conservatively directed, and the two modes produce the identical bound at the extremal index $N = 690988$.

The program was compiled against an independently installed FLINT/Arb 3.6.0 and run at 256-bit precision over the four configurations of (19), in each case at $y = y_0$ and over a closed rational t-interval containing t_0 (for the first range, the singleton $\{129/800\}$). Every one of the 3,149,013 values of N produced a certified positive bound, with no uncertified rows; the concatenated outputs cover $690988 \le N \le 3840000$ with no gaps or duplicates; and the certified per-range minima are as in Table 1, with global minimum 791366×10^{-12} at $N = 690988$. Per-range summaries (`runs/*_summary.txt`) and the complete per-N records (`runs/*_sweep.out`) are included in the supplementary material. □

4.4 Transfer to the full height range

Proposition 4.3 concerns the single height $y = y_0$, while hypothesis (ii) requires nonvanishing for all $y \in [y_0, \sqrt{1-2t_0}]$. For fixed N write $L_N(y)$ for the quantity (16) with its explicit y-dependence restored (the constants g_N, k_N, σ_N frozen in x as before, but evaluated at height y; likewise the coefficients $A_{N,n}$ and the sums M_N, C_N), so that $L_N = L_N(y_0)$. Throughout this subsection $t = t_0$ and N are fixed, and we abbreviate

$$q_N = N^2 - \frac{t_0}{16}, \qquad g = \frac{1}{50} - \frac{1}{2}\log q_N \ (<0), \qquad \rho_N = \frac{t_0}{2(x_N - 6)},$$

so that $g_N(y) = e^{yg}$ exactly and $k_N(y) = \rho_N y$. The bridge between Proposition 4.3 and the full height range is the following statement.

Proposition 4.4 (Height transfer)**.** *For every integer* $690988 \le N \le 3840000$ *and every* $y \in [y_0, \sqrt{1-2t_0}]$,

$$L_N(y) \ \ge \ L_N(y_0).$$

The proof establishes three monotonicity statements: as y increases over $[y_0, \sqrt{1-2t_0}]$, (a) the majorant sum

$$\mathcal{F}_{N,\mathcal{P}}(y) = g_N(y) + \sum_{n=2}^{DN} \big(|B_{N,n}| + g_N(y)|A_{N,n}(y)|\big) n^{-\sigma_N(y)}$$

subtracted in (16) is nonincreasing; (b) the mollifier bound $M_N(y)$ is positive and nonincreasing; and (c) the correction term $g_N(y)C_N(y) \ge 0$ is nonincreasing. Statement (a) is where an earlier, now-abandoned shortcut of Gomila's failed: the crude majorant obtained by discarding the negative term $g_N g|A|$ below is not monotone, and the proof must retain the cancellation between the A-side terms. We build it up in three lemmas and one certified computation.

Lemma 4.5 (Slope of the frozen exponent)**.** *For* $y_0 \le y_1 \le y_2 \le \sqrt{1-2t_0}$,

$$\sigma_N(y_2) - \sigma_N(y_1) \ \ge \ \tfrac{1}{2}(y_2 - y_1).$$

Proof. $\sigma_N(y) = \frac{1+y}{2} + \frac{t_0}{4}\log q_N - \frac{t_0}{2x_N^2}\big(1 - 3y + \frac{4y(1+y)}{x_N^2}\big)_+$. The inner expression has y-derivative $-3 + \frac{4(1+2y)}{x_N^2} < 0$ (as $4(1+2y) \le 12 < 3x_N^2$), so its positive part is nonincreasing in y and the subtracted term is nondecreasing; the explicit term $\frac{1+y}{2}$ contributes slope $\frac{1}{2}$. □

Lemma 4.6 (Monotone normalizer and correction)**.** *On* $[y_0, \sqrt{1-2t_0}]$*:* (b) $M_N(y)$ *is positive and nonincreasing;* (c) $g_N(y)C_N(y)$ *is nonnegative and nonincreasing.*

Proof. (b) Each term $|\lambda_d| d^{-\sigma_N(y)}$ is positive; for $d \geq 2$ it is nonincreasing by Lemma 4.5, and the $d = 1$ term is constant.

(c) Nonnegativity is clear ($m^{k_N(y)} \geq 1$). Fix $y_0 \leq y_1 < y_2 \leq \sqrt{1-2t_0}$ and put $\Delta = y_2 - y_1$. First, $g_N(y_2) = g_N(y_1)e^{g\Delta}$ exactly. Next, for each $2 \leq m \leq N$, the term $b_t(m)\,(m^{\rho_N y} - 1)\, m^{y-\sigma_N(y)}$ of C_N grows by at most the factor $e^{(\frac{1}{2}\log m + \rho_N \log m + \frac{1}{y_0})\Delta}$: the exponent $y - \sigma_N(y)$ grows by at most $\frac{1}{2}\Delta$ (Lemma 4.5); and by the mean value theorem applied to $y \mapsto \log(m^{\rho_N y} - 1)$, whose derivative is $\rho_N \log m\, \frac{e^z}{e^z - 1}$ with $z = \rho_N y \log m > 0$, the bound $\frac{e^z}{e^z-1} \leq 1 + \frac{1}{z}$ gives a growth rate at most $\rho_N \log m + \frac{1}{y} \leq \rho_N \log m + \frac{1}{y_0}$. Since a sum of positive terms grows no faster than its fastest-growing term, and $m \leq N$,

$$g_N(y_2)C_N(y_2) \;\leq\; g_N(y_1)C_N(y_1)\, \exp\Big[\Big(g + \tfrac{1}{2}\log N + \rho_N \log N + \tfrac{1}{y_0}\Big)\Delta\Big].$$

It remains to check that the rate is negative for every N in the range. First, $g + \frac{1}{2}\log N = \frac{1}{50} - \frac{1}{2}\log\frac{q_N}{N}$ and $\frac{q_N}{N} = N - \frac{t_0}{16N} > N - 1 \geq 690987 > e^{13.44}$ (as $e^{13.44} < 442414 \times 1.5528 < 687000$), so $g + \frac{1}{2}\log N < 0.02 - 6.72$. Second, $\frac{1}{y_0} < \frac{10^7}{1872719} < 5.3399$, since $y_0^2 = \frac{350708}{10^7} > \big(\frac{1872719}{10^7}\big)^2 = 0.03507076\ldots$ gives $y_0 > \frac{1872719}{10^7}$. Third, uniformly over the range, $\rho_N \log N \leq \frac{t_0 \log(3.84\times 10^6)}{2(x_{690988}-6)} < \frac{0.17\times 16}{10^{13}} < 10^{-12}$. Hence the rate is at most $0.02 + 5.3399 + 10^{-12} - 6.72 < -1.36 < 0$. □

Lemma 4.7 (Dini bound in divisor-cell form)**.** *Let $2 \leq n \leq DN$, $L = \log n$, and let $\mathcal{D}_{N,n} = \{d \mid D :\; d \mid n,\; n \leq dN\}$ be the active divisor set (which does not depend on y), assumed nonempty. With $\ell_d = \log d$ and*

$$\begin{aligned}
r_d(L) &= \exp\Big\{\frac{t_0}{4}\Big(\sum_{p\mid d}\log^2 p + \ell_d^2 - 2L\ell_d\Big)\Big\},\\
C_0 &= \sum_{d\in\mathcal{D}_{N,n}} (-1)^{\omega(d)} r_d(L),\\
C(y) &= \sum_{d\in\mathcal{D}_{N,n}} (-1)^{\omega(d)} r_d(L)\, d^{-y},\\
C'(y) &= -\sum_{d\in\mathcal{D}_{N,n}} (-1)^{\omega(d)} r_d(L)\, d^{-y}\ell_d,
\end{aligned}$$

one has $B_{N,n} = b_t(n)\,C_0$, $A_{N,n}(y) = b_t(n)\,e^{yL}C(y)$ and $\frac{d}{dy}A_{N,n}(y) = b_t(n)\,e^{yL}(LC(y) + C'(y))$, and the upper right Dini derivative of the corresponding summand of $\mathcal{F}_{N,\mathcal{P}}$ satisfies

$$D_y^+\Big[\big(|B_{N,n}| + g_N(y)|A_{N,n}(y)|\big)n^{-\sigma_N(y)}\Big] \;\leq\; b_t(n)\, n^{-\sigma_N(y)}\Big(G\,h - \frac{L}{2}\,|C_0|\Big),$$

where $G = G(y) = e^{y(g+L)}$ and $h = h(y) = |LC(y) + C'(y)| + \big(g - \frac{L}{2}\big)|C(y)|$.

Proof. The identities follow from $\lambda_d\, b_t(n/d) = (-1)^{\omega(d)} b_t(n)\, r_d(L)$ — indeed $\log b_t(n/d) = \frac{t_0}{4}(L - \ell_d)^2 = \log b_t(n) + \frac{t_0}{4}(\ell_d^2 - 2L\ell_d)$ and $\lambda_d = (-1)^{\omega(d)} \exp(\frac{t_0}{4}\sum_{p\mid d}\log^2 p)$ — together with $(n/d)^y = e^{yL}d^{-y}$; note also $g_N(y)e^{yL} = e^{y(g+L)} = G$. Since $A = A_{N,n}(y)$ is real-analytic in y,

$$D_y^+\{g_N|A|\} = \begin{cases} g_N\big(g|A| + \operatorname{sgn}(A)A'\big), & A \neq 0,\\ g_N|A'|, & A = 0,\end{cases} \;\leq\; g_N\big(g|A| + |A'|\big)$$

in both cases. Both factors of the summand are nonnegative and locally Lipschitz, $|B|$ is constant in y, and by Lemma 4.5, $D_y^+ n^{-\sigma_N(y)} \le -\frac{L}{2} n^{-\sigma_N(y)}$; the product rule for upper Dini derivatives of nonnegative Lipschitz functions then gives

$$D_y^+\big[(|B|+g_N|A|)n^{-\sigma_N}\big] \le n^{-\sigma_N}\Big(g_N\big(g|A|+|A'|\big) - \frac{L}{2}\big(|B|+g_N|A|\big)\Big).$$

Substituting the identities and factoring out $b_t(n) > 0$ yields the display. □

Lemma 4.8 (Cell decomposition and q-freezing)**.** (i) *For $2 \le n \le DN$ put $G_0 = \gcd(n, D)$ and list the divisors of G_0 as $1 = d_0 < d_1 < \cdots < d_s$ (so $s+1 = \tau(G_0)$), with $d_{-1} = 0$. On the ratio sector $d_{j-1} < n/N \le d_j$ $(0 \le j \le s)$ the active set is exactly $\mathcal{D}_{N,n} = \{d_j, \dots, d_s\}$, and for $n/N > d_s$ it is empty (the summand vanishes identically). The number of pairs (mask, sector) with nonempty active set is $\sum_{G_0|D} \tau(G_0) = 3^{|\mathcal{P}|}$.*

(ii) *With $c = |C(y)|$ and $K = |LC(y) + C'(y)| - \frac{L}{2}c$, so that $G\,h_+ = U(g) := e^{y(L+g)}\,[K + gc]_+$, the function $\tilde g \mapsto U(\tilde g)$ is nondecreasing.*

(iii) *On the sector of (i), within the range $N_{\min} \le N \le N_{\max}$ of the prime set, $q_N \ge \max\big(N_{\min}^2, e^{2L}/d_j^2\big) - \frac{t_0}{16}$; since g is decreasing in q_N, replacing g by its value at this smallest feasible q_N can only increase $U(g)$.*

Proof. (i) $d \in \mathcal{D}_{N,n}$ iff $d \mid G_0$ and $n \le dN$, i.e. $d \ge n/N$; on the stated sector these are exactly $d_j, \dots, d_s$. Summing $\tau(G_0)$ over the divisors G_0 of the squarefree D gives $\prod_{p\in\mathcal{P}}(\tau(1)+\tau(p)) = 3^{|\mathcal{P}|}$. (ii) Where $K + \tilde g c > 0$, $U'(\tilde g) = e^{y(L+\tilde g)}\big(y(K+\tilde g c) + c\big) \ge 0$; where $K + \tilde g c \le 0$, $U = 0$; and U is continuous. (iii) $n \le d_j N$ gives $N \ge e^L/d_j$, and $N \ge N_{\min}$; q_N is increasing in N and g decreasing in q_N. □

By Lemma 4.8, for each of the four prime sets the quantity $G\,h_+$ is bounded by an expression depending only on the cell, on L and on y; the following finite family of inequalities is therefore checkable by a finite computation.

Proposition 4.9 (Certified cell inequalities)**.** *For each of the four prime sets of (19), every N in the corresponding range, every $2 \le n \le DN$ with $\mathcal{D}_{N,n} \ne \emptyset$, and every $y \in [y_0, \sqrt{1-2t_0}]$, in the notation of Lemma 4.7,*

$$G\,h_+ \;\le\; \frac{L}{2}\,|C_0|.$$

Proof. The proof is computer-assisted; the verification program `prop49_proof.c`, included with this manuscript as supplementary material (directory `code/prop49`; derived from Gomila's source as documented in its header and reviewed line by line as part of this audit), reads no stored data and fails closed at every decision.

For each prime set, the program enumerates the $3^{|\mathcal{P}|}$ (mask, ratio-sector) cells of Lemma 4.8(i): for each divisor mask $G_0 \mid D$ and each divisor d_j of G_0, the active set $\{d_j, \dots, d_s\}$ with the sector constraint $d_{j-1} < n/N \le d_j$. The (L, y)-domain of the cell, namely

$$L \in [\max(\log 2, \log(N_{\min} d_{j-1})), \log(N_{\max} d_j)]$$

and $y \in [y_0, \sqrt{1-2t_0}]$, is covered by closed binary64 rectangles whose endpoints are padded outward by 2×10^{-12}; the containment of each exact logarithm in its padded double endpoint, and of the exact y_0 and $\sqrt{1-2t_0}$ in the fixed y-box, is itself certified by ball arithmetic before use. On each rectangle the program freezes q_N at the smallest feasible value

$\max\big(N_{\min}^2, e^{2L}/d_j^2\big) - \frac{t_0}{16}$ (splitting the L-range at the transition point with overlapping padding; conservative by Lemma 4.8(ii)–(iii)), forms ball enclosures of C_0, $C(y)$, $C'(y)$ from the exact identity $r_d(L) = e^{\frac{t_0}{4}(\sum_{p|d}\log^2 p+\ell_d^2)} e^{-\frac{t_0}{2}L\ell_d}$, and certifies either $h \le 0$ or the strict ball inequality $G\,h_+ < \frac{L}{2}|C_0|$, bisecting the longer side adaptively (depth at most 42) and failing closed on any unresolved leaf. Before starting, the program also verifies numerically, at the smallest window, the two elementary inequalities $4(1+2y) < 3x_{690988}^2$ and $g < 0$: the first is the assumption underlying Lemma 4.5, and the second makes the standalone term $g_N(y) = e^{yg}$ decreasing in the proof of Proposition 4.4. Both have already been verified analytically in the proofs just cited, and the numerical check adds nothing to the present proof; it is a deliberate redundancy of Gomila's program, which refuses to run when these inequalities fail, so that a passing run remains meaningful even if the parameters hard-coded in the program were ever modified.

The fresh runs, at both 180- and 256-bit precision, certify all four prime sets with no unresolved leaves (297490, 52239, 9290 and 237 leaves respectively, identical covers at both precisions), and the aggregated certified ratio bound

$$\sup \frac{G\,h_+}{\frac{L}{2}|C_0|} \;\le\; 0.99999860767275095 \;<\; 0.9999988 \;<\; 1$$

(the closing comparison against the fixed threshold 0.9999988 was added in this review; Gomila's original certified only the per-leaf inequalities and printed the worst ratio without checking it). The worst leaf lies in the five-prime range at $L \approx 1.042$–1.054, $y \approx 0.1997$–0.2022; its margin $1 - \text{ratio} \approx 1.4 \times 10^{-6}$ is the second-thinnest in the entire proof. Each run completes in under ten seconds; the outputs are included as `code/prop49/runs/prop49_report_180.txt` and `prop49_report_256.txt`. □

Proof of Proposition 4.4. Statement (a). For each n with nonempty active set, Lemma 4.7 and Proposition 4.9 give $D_y^+ \le b_t(n) n^{-\sigma_N}(G\,h - \frac{L}{2}|C_0|) \le 0$ on $[y_0, \sqrt{1-2t_0}]$ (if $h \le 0$ both terms are nonpositive; otherwise apply the proposition). Summands with empty active set vanish identically, and the standalone term $g_N(y) = e^{yg}$ is decreasing. Each summand is continuous, and a continuous function with nonpositive upper right Dini derivative on an interval is nonincreasing (if $f(y_2) > f(y_1)$, put $\varepsilon = \frac{f(y_2)-f(y_1)}{2(y_2-y_1)}$ and let y^* be the supremum of the set where $f(y) \le f(y_1) + \varepsilon(y - y_1)$; then $y^* < y_2$ and $D^+ f(y^*) \ge \varepsilon > 0$). Hence $\mathcal{F}_{N,\mathcal{P}}$ is nonincreasing.

Conclusion. By Proposition 4.3, $1 - \mathcal{F}_{N,\mathcal{P}}(y_0) = M_N(y_0)L_N(y_0) + M_N(y_0)\,g_N(y_0)C_N(y_0) \ge M_N(y_0)\,T_{\min} > 0$. By (a) the numerator $1 - \mathcal{F}_{N,\mathcal{P}}(y)$ is nondecreasing, hence positive on the whole interval; by Lemma 4.6(b) the denominator $M_N(y)$ is positive and nonincreasing; so the quotient $(1 - \mathcal{F}_{N,\mathcal{P}}(y))/M_N(y)$ is nondecreasing. Subtracting the nonnegative nonincreasing correction $g_N(y)C_N(y)$ (Lemma 4.6(c)) preserves this, and $L_N(y) \ge L_N(y_0)$ follows. □

4.5 Error bounds and the finite-range conclusion

The remaining ingredient is a uniform bound on the errors of Theorem 1.3 over the finite region.

Proposition 4.10 (Uniform error bound, finite region)**.** *For all $x \ge x_*$ and $y_0 \le y \le \sqrt{1-2t_0}$, the error terms of Theorem 1.3 at $t = t_0$, bounded using equation (23) of [7] for $e_A + e_B$ and the conservative corollary (13) for $e_{C,0}$, satisfy*

$$e_A + e_B \le 2.06 \times 10^{-12}, \qquad e_{C,0} \le 0.000000233492848188649183,$$

and

$$e_A + e_B + e_{C,0} \ \leq \ E_{\max} := 0.000000233494905212337849.$$

(The third bound is certified directly and is sharper than the sum of the two displayed bounds, which are rounded upward for display.)

Proof. The proof is computer-assisted; the verification program `prop410_proof.py`, included with this manuscript as supplementary material (directory `code/prop410`; derived from Gomila's source as documented in its header and reviewed line by line as part of this audit), computes outward-rounded interval enclosures (220-bit precision) of majorants of the right-hand sides of equation (23) of [7] and of the corollary (13). Its only inputs are the exact rational parameters t_0, y_0^2 and $N_0 = 690988$ (the enclosures are computed over the rational interval $[t_0,\, t_0 + 10^{-9}]$, which contains t_0); it reads no stored data, and every acceptance condition below is a strict inequality between exact rational endpoints, checked in exact arithmetic.

A single evaluation dominates the whole region $x \geq x_*$, $y_0 \leq y \leq \sqrt{1-2t_0}$, $t = t_0$ by the following monotonicity reductions, whose hypotheses are among the twelve domain and sign conditions certified by the program. Write $\ell = \log \frac{x_*}{4\pi} = \log\bigl(N_0^2 - \frac{t_0}{16}\bigr)$,

$$\delta_1 = \frac{t_0}{4}\log\Bigl(1 - \frac{t_0}{16N_0^2}\Bigr)^{-1} + \frac{t_0}{2x_*^2}, \qquad \sigma_1 = \frac{1+y_0}{2} + \frac{t_0}{2}\log N_0 - \delta_1 = \frac{1+y_0}{2} + \frac{t_0}{4}\ell - \frac{t_0}{2x_*^2}.$$

(i) *The exponent.* In (11) the positive part is at most 1: the program certifies $8/x_*^2 < 3y_0$, so $\frac{4y(1+y)}{x^2} \leq \frac{8y}{x^2} < 3y$ for all $y \geq y_0$, and $\operatorname{Re} s_* \geq \frac{1+y}{2} + \frac{t_0}{4}\log\frac{x}{4\pi} - \frac{t_0}{2x^2} \geq \sigma_1$ throughout the region.

(ii) *The bound* (13) *for* $e_{C,0}$. Each factor is monotone in the favorable direction: the prefactor $(\frac{x}{4\pi})^{-(1+y)/4}$ and the term $-\frac{t_0}{16}\log^2\frac{x}{4\pi}$ decrease in x (certified: $\ell > 2 > 0$) and in y; the factor $3^y + 3^{-y}$ increases in y; $N - 0.125$ increases with the window; and the final quotient decreases in x because $3\,\frac{d}{dx}\bigl|\log\frac{x}{4\pi} + \frac{i\pi}{2}\bigr| \leq \frac{3}{x}$ while the numerator exceeds $10.50 > 3$ (certified). Hence (13) evaluated at $x = x_*$ and $N = N_0$, with $y = y_0$ in the decreasing factors and $y = \sqrt{1-2t_0}$ in $3^y + 3^{-y}$, bounds $e_{C,0}$ on the whole region; this evaluation is the program's quantity `error_c0`.

(iii) *The bound (23) for* $e_A + e_B$. Since $4\pi n^2 \leq 4\pi N^2 \leq x\,(1 - \frac{t_0}{16N^2})^{-1}$ for every $n \leq N$, the bounds (10) and (12) give, uniformly on the region,

$$1 + |\gamma| N^{|\kappa|} n^y \ \leq \ 1 + e^{0.02}\Bigl(1 - \frac{t_0}{16N_0^2}\Bigr)^{-1/2}\exp\Bigl(\frac{t_0\log N_0}{2(x_*-6)}\Bigr)$$

(the quantity $\log N/(x_N - 6)$ decreases in N), while $\log^2\frac{x}{4\pi n^2} \leq \log^2\frac{x}{4\pi}$ for $n \leq N$ and the monotonicity of $\log^2(\frac{x}{4\pi})/(x - 6.66)$ in x (certified: $\ell > 2$) bound the last factor of (23) by $\exp\bigl(\frac{(t_0^2/16)\ell^2 + 0.626}{x_* - 6.66}\bigr) - 1$. For the middle factor, $\sum_{n=1}^{N} b_{t_0}(n)\, n^{-\operatorname{Re} s_*} \leq 1 + \sum_{n=2}^{m_0} b_{t_0}(n)\, n^{-\sigma_1} + T_N$ with $m_0 = 2000$, where T_N is the sum over $m_0 < n \leq N$. The summand $\exp(\frac{t_0}{4}\log^2 n - \sigma_N \log n)$ decreases in n on $n \leq N$ (certified: $\frac{1+y_0}{2} - \delta_1 > 0$), so $T_N \leq \int_{\log m_0}^{\log N} e^{\psi(L)}\,dL$ with $\psi(L) = (1-\sigma_N)L + \frac{t_0}{4}L^2$; since ψ is convex, $T_N \leq (\log N - \log m_0)\,\max\bigl(e^{\psi(\log m_0)}, e^{\psi(\log N)}\bigr)$. Finally, the two certified rate inequalities

$$\frac{t_0}{2}\log m_0 \ > \ \Bigl(\log\frac{N_0}{m_0}\Bigr)^{-1}, \qquad \frac{t_0}{2}\log N_0 - \frac{1-y_0}{2} - \delta_1 \ > \ \Bigl(\log\frac{N_0}{m_0}\Bigr)^{-1},$$

imply that each of the two endpoint products $(\log N - \log m_0)e^{\psi(\log m_0)}$ and $(\log N - \log m_0)e^{\psi(\log N)}$ is nonincreasing in N for $N \geq N_0$ (using that σ_N grows in $\log N$ at rate at least $\frac{t_0}{2}$, certified via $N_0^2/(N_0^2 - \frac{t_0}{16}) > 1$, and that $\delta_N \leq \delta_1$), so their values at $N = N_0$ dominate every window. The product of the three bounds is the program's quantity `error_ab`.

The program certifies the twelve domain and sign conditions required above, computes the enclosures of `error_ab`, `error_c0` and of their interval sum, and establishes the three displayed inequalities by exact rational comparison of the outward-rounded upper endpoints against the displayed constants. The run completes in under a second; a human-readable report of all intermediate enclosures is written to `code/prop410/runs/prop410_report.txt`. □

We can now state the conclusion of the finite range.

Proposition 4.11 (Nonvanishing at time t_0, finite range). *$H_{t_0}(x + iy) \neq 0$ for all $x_* \leq x < x_{3840001}$ and $y_0 \leq y \leq \sqrt{1 - 2t_0}$.*

Proof. Let x be in the stated range and let N be its window index, so that $690988 \leq N \leq 3840000$ by (15), and $x \in W_N$. For y in the stated range, Lemma 4.2 (applied with the height y, the prime set $\mathcal{P}(N)$, and the frozen constants of Lemma 4.1), together with Propositions 4.3 and 4.4, gives

$$|f_{t_0}(x + iy)| \;\geq\; L_N(y) \;\geq\; L_N(y_0) \;\geq\; T_{\min}.$$

Combining with Proposition 4.10,

$$|f_{t_0}(x + iy)| - \big(e_A + e_B + e_{C,0}\big) \;\geq\; T_{\min} - E_{\max} \;\geq\; 0.000000557871094787 \;>\; 0. \tag{20}$$

The point (x, y, t_0) lies in the region (7), so [7, Corollary 1.4] (the criterion for non-vanishing quoted in Section 2.2) yields $H_{t_0}(x + iy) \neq 0$. □

5 Hypothesis (ii), infinite tail

This section proves nonvanishing of H_t for *all* windows $N \geq N_* = 3840000$, all $t \in I_t = \left[\frac{129}{800}, \frac{161250001}{10^9}\right]$ (a closed interval whose left endpoint is exactly t_0), and all $y \in [y_0, \sqrt{1 - 2t}]$, by a single interval computation at the cutoff N_* combined with monotonicity arguments in N and y. Since $t_0 \in I_t$, this in particular covers the case $t = t_0$ needed for hypothesis (ii); the last window $W_{3840000}$ of the finite range overlaps the tail's starting abscissa $x_{3840000}$, so the two ranges jointly cover all $x \geq x_*$.

5.1 Setup: mollifier and frozen exponents

Fix the sparse mollifier support $\mathcal{S} = \{1, 2, 3, 4, 5, 6, 7, 10, 11, 13, 14\}$ and the exact rational vector

d	1	2	3	4	5	6	7	10	11	13	14
λ_d	1	$-\frac{1021}{1000}$	$-\frac{1054}{1000}$	$-\frac{9}{200}$	$-\frac{1119}{1000}$	$\frac{1001}{1000}$	$-\frac{1043}{1000}$	$\frac{128}{125}$	$-\frac{161}{200}$	$-\frac{447}{500}$	$\frac{456373}{500000}$

($\lambda_1 = 1$; $\sum_d |\lambda_d| = 9.918746$ exactly), and set $M_\lambda(z) = \sum_{d \in \mathcal{S}} \lambda_d d^{-z}$. (Unlike the Euler-product mollifier of Section 4.2, this vector was chosen by numerical optimization; its only structural requirement is $\lambda_1 = 1$ and real entries.) Fix also the head lengths $M = 153814$ and $M_{\text{err}} = 3000$.

For $x \geq X_N(t) := 4\pi\bigl(N^2 - \frac{t}{16}\bigr)$ with window index $N \geq N_*$, define frozen exponents (uniform over I_t, via rigorous upper bounds $\widehat{\delta} \geq \delta(N_*, t)$, $\widehat{k} \geq k(N_*, t)$ for the quantities $\delta(N,t) = \frac{t}{4}\bigl[-\log\bigl(1 - \frac{t}{16N^2}\bigr)\bigr] + \frac{t}{2X_N(t)^2}$ and $k(N,t) = \frac{t}{2(X_N(t)-6)}$, both decreasing in N):

$$\sigma_1 = \frac{1+y}{2} + \frac{t}{2}\log N - \widehat{\delta}, \qquad \sigma_2 = \frac{1-y}{2} + \frac{t}{2}\log N - \widehat{\delta} - \widehat{k}, \qquad G = e^{0.02y}\Bigl(N^2 - \frac{t}{16}\Bigr)^{-y/2}.$$

Lemma 5.1 (Frozen exponents)**.** *Let $N \geq N_*$, $t \in I_t$, $0 < y \leq 1$, and let $x \in [X_N(t), X_{N+1}(t))$, so that x has window index N. Then*

$$|\gamma| \leq G, \qquad \operatorname{Re} s_* \geq \sigma_1, \qquad |\kappa| \leq \widehat{k},$$

and consequently each term of the second sum in (9) satisfies

$$\bigl|n^y\, b_t(n)\, n^{-\overline{s_*}-\kappa}\bigr| \;\leq\; b_t(n)\, n^{-\sigma_2} \qquad (1 \leq n \leq N).$$

Proof. Since $x \geq X_N(t)$ we have $x/4\pi \geq N^2 - t/16$, and the three factors $(x/4\pi)^{-y/2}$, x^{-2}, $(x-6)^{-1}$ appearing in (10)–(12) are decreasing in x, hence maximal at the window's left edge. For γ this gives $|\gamma| \leq e^{0.02y}(N^2 - t/16)^{-y/2} = G$ directly. For κ, using $y \leq 1$ and that $X_N(t)$ increases with N,

$$|\kappa| \;\leq\; \frac{ty}{2(x-6)} \;\leq\; \frac{t}{2(X_N(t)-6)} \;=\; k(N,t) \;\leq\; k(N_*,t) \;\leq\; \widehat{k}.$$

For $\operatorname{Re} s_*$: by the exact identity

$$\frac{t}{4}\log\Bigl(N^2 - \frac{t}{16}\Bigr) = \frac{t}{2}\log N - \frac{t}{4}\Bigl[-\log\Bigl(1 - \frac{t}{16N^2}\Bigr)\Bigr],$$

the logarithmic term of (11) is at least $\frac{t}{2}\log N$ minus the first summand of $\delta(N,t)$. The subtracted positive-part term of (11) is at most $t/(2x^2) \leq t/(2X_N(t)^2)$, the second summand of $\delta(N,t)$: indeed, on the domain $0 \leq y \leq 1$, $x \geq 200$ one has $1 - 3y + 4y(1+y)/x^2 \leq 1$ (for $y > 0$ this is $4(1+y) \leq 3x^2$, and at $y = 0$ the left side equals 1), so the positive part is at most 1. Both summands of $\delta(N,t)$ decrease in N (in the first, $t/16N^2$ decreases and $u \mapsto -\log(1-u)$ is increasing; in the second, $X_N(t)$ increases), so

$$\operatorname{Re} s_* \;\geq\; \frac{1+y}{2} + \frac{t}{2}\log N - \delta(N,t) \;\geq\; \frac{1+y}{2} + \frac{t}{2}\log N - \widehat{\delta} \;=\; \sigma_1.$$

Finally, for $1 \leq n \leq N$,

$$\bigl|n^y b_t(n) n^{-\overline{s_*}-\kappa}\bigr| = b_t(n)\, n^{\,y - \operatorname{Re} s_* - \operatorname{Re}\kappa} \leq b_t(n)\, n^{\,y-\sigma_1+\widehat{k}} = b_t(n)\, n^{-\sigma_2},$$

since $-\operatorname{Re}\kappa \leq |\kappa| \leq \widehat{k}$ and $\sigma_2 = \sigma_1 - y - \widehat{k}$. $\square$

5.2 The endpoint-cap lemma

The analytic engine for tail sums is:

Lemma 5.2 (Endpoint cap)**.** *For* $1 \le a < c$ *set* $E_{t,\sigma}(u) = \exp\big((1-\sigma)\log u + \frac{t}{4}\log^2 u\big)$ *and*

$$\mathrm{Cap}_t(a,c;\sigma) = \max\big(E_{t,\sigma}(a), E_{t,\sigma}(c)\big)\log\frac{c}{a}.$$

If $\sigma > \frac{t}{2}\log c$*, then for all integers* $a \le J < K \le c$*,*

$$\sum_{J<n\le K} b_t(n)\, n^{-\sigma} \;\le\; \mathrm{Cap}_t(a,c;\sigma).$$

Proof. The logarithmic derivative of $b_t(u)u^{-\sigma}$ in $\log u$ is $\frac{t}{2}\log u - \sigma < 0$ on $[a,c]$, so the summand decreases and the sum is at most $\int_a^c b_t(u)u^{-\sigma}du$. Substituting $v = \log u$ makes the integrand $\exp\big((1-\sigma)v + \frac{t}{4}v^2\big)$, which is log-convex in v, hence maximized at an endpoint of $[\log a, \log c]$; multiplying by the length $\log(c/a)$ of the v-interval gives the bound. $\square$

5.3 Exact convolution and the contraction D

Define, for $N \ge N_*$ and (x,y,t) with x in the window of index N,

$$P = \sum_{m=2}^{M} |c_m(t)| m^{-\sigma_1}, \qquad c_m(t) = \sum_{\substack{d\in\mathcal{S}\\ d\mid m}} \lambda_d\, b_t(m/d), \qquad M_{\max} = \sum_{d\in\mathcal{S}} |\lambda_d|\, d^{-\sigma_1},$$

$$\mathrm{TR} = \sum_{d\in\mathcal{S}} |\lambda_d|\, d^{-\sigma_1}\,\mathrm{Cap}_t\big(\lfloor M/d\rfloor, N; \sigma_1\big), \qquad \mathrm{AB} = G\Big(\sum_{k\le M} b_t(k)k^{-\sigma_2} + \mathrm{Cap}_t(M,N;\sigma_2)\Big),$$

together with the nonnegative "overshoot" term

$$\mathrm{OV} = \sum_{\substack{d\in\mathcal{S}\\ d>1}} |\lambda_d|\, d^{-\sigma_1}\;\mathrm{Cap}_t\Big(\frac{N}{d+1},\, N; \sigma_1\Big)$$

(included by Gomila's implementation; see the remark after the proof), and set

$$D \;=\; D(N,t,y) \;:=\; P + \mathrm{TR} + \mathrm{OV} + M_{\max}\,\mathrm{AB}. \tag{21}$$

Lemma 5.3 (Mollified contraction bound)**.** *Assume the cap-validity conditions*

$$\widehat{\delta} \;<\; \frac{1+y}{2}, \qquad \widehat{\delta} + \widehat{k} \;<\; \frac{1-y}{2} \tag{22}$$

(equivalently $\sigma_1 > \frac{t}{2}\log N$ *and* $\sigma_2 > \frac{t}{2}\log N$*: the differences* $\sigma_i - \frac{t}{2}\log N$ *are independent of* N*). Then for every* $N \ge N_*$*,* $t \in I_t$*,* $0 < y \le 1$ *and* x *in the window of index* N*,*

$$\big|M_\lambda(s_*) f_t(x+iy) - 1\big| \;\le\; D.$$

Consequently, if $D < 1$ *then*

$$|f_t(x+iy)| \;\ge\; \frac{1-D}{M_{\max}}. \tag{23}$$

Proof. Write the two sums in (9) as F_B and F_A ($f_t = F_B + \gamma F_A$). Expanding $M_\lambda(s_*)F_B = \sum_{d\in\mathcal{S}}\sum_{k\le N}\lambda_d\, b_t(k)\,(dk)^{-s_*}$ and partitioning the index pairs (d,k) by $dk \le M$ versus $dk > M$ — a disjoint and exhaustive partition — gives the *exact* identity

$$M_\lambda(s_*)F_B - 1 = \sum_{m=2}^{M} c_m(t)\, m^{-s_*} + \sum_{d\in\mathcal{S}} \lambda_d\, d^{-s_*} \sum_{\lfloor M/d\rfloor < k \le N} b_t(k)\, k^{-s_*}. \tag{24}$$

Here the first term groups the pairs with $dk \le M$ by $m = dk$: since $m \le M < N_* \le N$, the constraint $k = m/d \le N$ is automatic, so the coefficient of m^{-s_*} is exactly $c_m(t)$, and the $m = 1$ term $c_1 = \lambda_1 b_t(1) = 1$ has been moved to the left side. In the second term, for fixed d the condition $dk > M$ for integer k is $k > \lfloor M/d\rfloor$. By $\operatorname{Re} s_* \ge \sigma_1$ (Lemma 5.1), the first term is bounded by P; and each inner sum of the second term is bounded, by Lemma 5.2 with $(a,c) = (\lfloor M/d\rfloor, N)$, by $\mathrm{Cap}_t(\lfloor M/d\rfloor, N;\sigma_1)$, so the second term is bounded by TR; the application of Lemma 5.2 is legitimate because its hypothesis $\sigma_1 > \frac{t}{2}\log N$ is the first condition of (22). On the A-side, Lemma 5.1 bounds each term of $|\gamma F_A|$ by $G\, b_t(k)k^{-\sigma_2}$; splitting at $k = M$ and applying Lemma 5.2 with $(a,c) = (M,N)$ at exponent σ_2 (legitimate by the second condition of (22)) gives $|\gamma F_A| \le \mathrm{AB}$, while $|M_\lambda(s_*)| \le \sum_d |\lambda_d| d^{-\sigma_1} = M_{\max}$; hence $|M_\lambda(s_*)\gamma F_A| \le M_{\max}\mathrm{AB}$. Adding the nonnegative OV only weakens the bound, and the first claim follows. For the second claim: if $D < 1$ then $|M_\lambda(s_*)f_t| \ge 1 - D > 0$; in particular $M_\lambda(s_*) \ne 0$, and dividing by $|M_\lambda(s_*)| \le M_{\max}$ gives $|f_t| \ge (1-D)/M_{\max}$. □

Remark 5.4. The term OV plays no role in the mathematics — the identity (24) is fully covered by $P + \mathrm{TR}$ — but Gomila's program includes it in D, so it is included in (21) in order that the certified numerical bound on D (Proposition 5.10 below) majorizes the same quantity the program computes.

5.4 Reduction to the cutoff

The interval computation of Section 5.6 evaluates D, and the error majorants, only at the cutoff $N = N_*$, over the closed intervals

$$I_{\text{box}} = \left[\tfrac{1872719}{10^7}, \tfrac{23409}{125000}\right], \qquad I_{\text{ext}} = \left[\tfrac{1872719}{10^7}, \tfrac{8231039}{10^7}\right],$$

where I_{box} contains y_0 and the top of I_{ext} is at least $\sqrt{1-2t}$ for every $t \in I_t$. The reduction from this single evaluation to all $N \ge N_*$ and the full height range is carried out in Lemmas 5.5–5.9 below. Throughout, the *cutoff conditions* are the following inequalities, all evaluated at $N = N_*$, uniformly for $t \in I_t$:

$$\frac{t}{2}\log a\,(\log N_* - \log a) > 1 \quad \text{and} \quad (\sigma - 1)(\log N_* - \log a) > 1 \tag{25}$$

for each cap $\mathrm{Cap}_t(a,\cdot\,;\sigma)$ appearing in D and in the error bounds (the left endpoints $a = \lfloor M/d\rfloor$ for $d \in \mathcal{S}$, $a = M$, and $a = M_{\text{err}}$, with $\sigma = \sigma_1$ or σ_2 as appropriate), together with the A-side slope condition

$$0.02 - \tfrac{1}{2}\log\!\left(N_*^2 - \tfrac{t}{16}\right) + \tfrac{1}{2}\log N_* \;<\; 0. \tag{26}$$

Lemma 5.5 (Decay of fixed-endpoint caps)**.** *Let $t > 0$, let $1 < a < N_*$, let $\sigma(q) = \sigma_0 + \frac{t}{2}q$ with $\sigma_0 \in \mathbb{R}$, and write $q_* = \log N_*$. Assume the two conditions (25) for this pair, that is,*

$\frac{t}{2}\log a\,(q_* - \log a) > 1$ and $(\sigma(q_*) - 1)(q_* - \log a) > 1$. Then both conditions persist for all $q \ge q_*$, one has $\sigma(q) > 1$ for all $q \ge q_*$, and

$$q \longmapsto \mathrm{Cap}_t\big(a, e^q; \sigma(q)\big)$$

is positive and decreasing on $[q_*, \infty)$.

Proof. By definition $\mathrm{Cap}_t(a, e^q; \sigma(q)) = \max\big(F_1(q), F_2(q)\big)$, where

$$F_1(q) = E_{t,\sigma(q)}(a)\,(q - \log a), \qquad F_2(q) = E_{t,\sigma(q)}(e^q)\,(q - \log a).$$

Since $\log E_{t,\sigma}(u) = (1-\sigma)\log u + \frac{t}{4}\log^2 u$ and $\sigma'(q) = \frac{t}{2}$,

$$\frac{d}{dq}\log F_1(q) = -\frac{t}{2}\log a + \frac{1}{q - \log a},$$

which is negative exactly when $\frac{t}{2}\log a\,(q - \log a) > 1$; both factors on the left of this condition are positive and nondecreasing in q, so the condition, true at q_*, persists for all $q \ge q_*$. Similarly,

$$\frac{d}{dq}\log F_2(q) = \Big[(1 - \sigma(q)) - \frac{t}{2}\,q\Big] + \frac{t}{2}\,q + \frac{1}{q - \log a} = 1 - \sigma(q) + \frac{1}{q - \log a},$$

which is negative exactly when $(\sigma(q)-1)(q - \log a) > 1$. At q_* this condition forces $\sigma(q_*) > 1$; then both factors $\sigma(q) - 1$ and $q - \log a$ are positive and increasing in q, so the condition persists, and $\sigma(q) > 1$ throughout. Hence F_1 and F_2 are positive and decreasing on $[q_*, \infty)$, and so is their maximum. □

Lemma 5.6 (Componentwise decay in N)**.** *Fix $t \in I_t$ and $y \in (0, 1]$, and assume the cutoff conditions (25). Then each of the following is nonincreasing in N on $[N_*, \infty)$ (throughout, $q = \log N$ and the frozen exponents $\sigma_i = \sigma_i(q)$ are increasing in q with slope $\frac{t}{2}$):*

(i) *$m^{-\sigma_i(q)}$ for every fixed integer $m \ge 2$ and $i \in \{1, 2\}$; hence the head P, the mollifier bound $M_{\max}$, and the A-side head $\sum_{k \le M} b_t(k)k^{-\sigma_2}$ (whose $k = 1$ term is constant) are nonincreasing;*

(ii) *$G = e^{0.02y}\big(e^{2q} - \frac{t}{16}\big)^{-y/2}$;*

(iii) *the moving-endpoint term* OV*.*

Proof. (i) is immediate: $\sigma_i(q)$ is increasing and $\log m > 0$, while the coefficients $|c_m(t)|$, $|\lambda_d|$ and $b_t(k)$ do not depend on N. For (ii),

$$\frac{d}{dq}\log G = -\frac{y}{2}\cdot\frac{2e^{2q}}{e^{2q} - t/16} = -\,y\,\frac{N^2}{N^2 - t/16} \;<\; 0.$$

For (iii), each summand of OV is $|\lambda_d|\,d^{-\sigma_1(q)}\,\mathrm{Cap}_t(e^q/(d+1), e^q; \sigma_1(q))$, a product of the nonincreasing factor $d^{-\sigma_1(q)}$ $(d \ge 2)$ with a cap whose endpoints both scale with N and whose length $\log(d+1)$ is constant. For $r \in \{1, d+1\}$,

$$\frac{d}{dq}\Big[(1-\sigma(q))(q-\log r) + \frac{t}{4}(q-\log r)^2\Big] = (1-\sigma(q)) - \frac{t}{2}(q-\log r) + \frac{t}{2}(q-\log r) = 1-\sigma(q) \;<\; 0,$$

using $\sigma_1(q) > 1$ from Lemma 5.5. So both cap candidates are positive and decreasing, hence so are their maximum and the summand. □

Lemma 5.7 (Decay in y at the cutoff)**.** *Fix $t \in I_t$ and assume the A-side slope condition* (26)*. Then $y \mapsto D(N_*, t, y)$ and $y \mapsto M_{\max}(N_*, t, y)$ are nonincreasing on* $(0, 1]$.

Proof. Recall $\partial\sigma_1/\partial y = \frac{1}{2}$ and $\partial\sigma_2/\partial y = -\frac{1}{2}$. On the B-side, every y-dependent factor has the form $u^{-\sigma_1(y)}$ or $E_{t,\sigma_1(y)}(u)$ with $u > 1$ (the head terms $m \geq 2$, the mollifier terms $d \geq 2$, and the cap ingredients at $u = \lfloor M/d \rfloor$, $u = N_*$ or $u = N_*/r$, all exceeding 1), and satisfies $\partial_y \log(\cdot) = -\frac{1}{2}\log u < 0$; the remaining factors ($|c_m(t)|$, $|\lambda_d|$, b_t, cap lengths) are independent of y, and maxima of nonincreasing functions are nonincreasing. Hence P, TR, OV and $M_{\max}$ are nonincreasing in y. On the A-side, $\log G$ is linear in y with slope $0.02 - \frac{1}{2}\log\bigl(N_*^2 - \frac{t}{16}\bigr)$, so each head term and each cap ingredient of AB, with $u \leq N_*$, satisfies

$$\partial_y \log\Bigl(G\, b_t(u)\, u^{-\sigma_2(y)}\Bigr) \text{ resp. } \partial_y \log\Bigl(G\, E_{t,\sigma_2(y)}(u)\Bigr) \;=\; 0.02 - \tfrac{1}{2}\log\Bigl(N_*^2 - \tfrac{t}{16}\Bigr) + \tfrac{1}{2}\log u,$$

which is at most the left side of (26), hence negative. So AB is nonincreasing in y, and therefore so are $M_{\max}$AB and D. ☐

Lemma 5.8 (Reduction to the cutoff)**.** *Assume the cutoff conditions* (25)*–*(26) *hold. Then for every $N \geq N_*$, every $t \in I_t$ and every $y \in [y_0, \sqrt{1-2t}]$,*

$$D(N,t,y) \;\leq\; \sup_{y' \in I_{\text{box}}} D(N_*, t, y'), \qquad M_{\max}(N,t,y) \;\leq\; \sup_{y' \in I_{\text{box}}} M_{\max}(N_*, t, y').$$

Proof. Fix t and y. Every constituent of $D = P + \text{TR} + \text{OV} + M_{\max}\text{AB}$ is a sum of products of nonnegative factors, each nonincreasing in N on $[N_*, \infty)$: the heads, $M_{\max}$ and G by Lemma 5.6(i)–(ii); the fixed-endpoint caps in TR and AB by Lemma 5.5, applied with $a = \lfloor M/d \rfloor$ (at σ_1) and $a = M$ (at σ_2), whose conditions (25) hold by hypothesis; and OV by Lemma 5.6(iii). Sums and products of nonnegative nonincreasing functions are nonincreasing, so $D(N,t,y) \leq D(N_*,t,y)$ and $M_{\max}(N,t,y) \leq M_{\max}(N_*,t,y)$. If $y \in I_{\text{box}}$ the claims follow at once. Otherwise $y > \max I_{\text{box}}$, since $y \geq y_0 \geq \min I_{\text{box}}$; as $y \leq \sqrt{1-2t} < 1$, Lemma 5.7 applies on $[\max I_{\text{box}}, y]$ and gives $D(N_*,t,y) \leq D(N_*,t,\max I_{\text{box}}) \leq \sup_{y' \in I_{\text{box}}} D(N_*,t,y')$, and the same for $M_{\max}$. ☐

5.5 Tail error terms

The error bounds use the exact definitions (71)–(72) of [7] rather than the coarser display (23): (23) applies the worst-case factor $N^{|\kappa|}$ to every term, whereas (71) carries the per-term factor $n^{-\Re\kappa}$, which Lemma 5.1 bounds by $n^{\widehat{k}}$. Define, for $N \geq N_*$, $t \in I_t$ and $0 < y \leq 1$,

$$L_N(t) = \log\Bigl(N^2 - \frac{t}{16}\Bigr), \qquad \Delta(N,t) = \frac{\frac{t^2}{16} L_N(t)^2 + 0.626}{X_N(t) - 6.66},$$

and the two majorants

$$\begin{aligned}
\mathrm{E}_{AB}(N,t,y) &= \bigl(e^{\Delta(N,t)} - 1\bigr)\bigl(S_1(N,t,y) + G\, S_2(N,t,y)\bigr),\\
S_i(N,t,y) &= \sum_{n=1}^{M_{\text{err}}} b_t(n)\, n^{-\sigma_i} + \mathrm{Cap}_t\bigl(M_{\text{err}}, N; \sigma_i\bigr) \qquad (i = 1,2),\\
\mathrm{E}_C(N,t,y) &= \Bigl(N^2 - \frac{t}{16}\Bigr)^{-(1+y)/4} \exp\Bigl(-\frac{t}{16} L_N(t)^2 + \frac{1.24\,(3^y + 3^{-y})}{N - 0.125} + \frac{3\sqrt{L_N(t)^2 + \pi^2/4} + 10.50}{X_N(t) - 12}\Bigr).
\end{aligned}$$

Lemma 5.9 (Error majorants)**.** *Assume the cap-validity conditions* (22) *and the conditions* (25) *for the caps with left endpoint* $a = M_{\mathrm{err}}$. *Then for every* $N \geq N_*$, $t \in I_t$, $y \in [y_0, \sqrt{1-2t}]$ *and* x *in the window of index* N,

$$e_A + e_B \;\leq\; \mathrm{E}_{AB}(N,t,y) \;\leq\; \mathrm{E}_{AB}(N_*,t,y), \qquad e_{C,0} \;\leq\; \mathrm{E}_C(N,t,y) \;\leq\; \mathrm{E}_C(N_*,t,y).$$

Proof. The per-term factor. The quantity $\varepsilon_{t,n}$ of [7, (44)] is bounded, by the display in the proof of [7, Proposition 6.6(iv),(v)],

$$\varepsilon_{t,n}\Big(\frac{1 \pm y + ix}{2}\Big) \;\leq\; \exp\Bigg(\frac{\frac{t^2}{32}\log^2\frac{x}{4\pi n^2} + 0.313}{\frac{x}{2} - 3.33}\Bigg) - 1 \;=\; \exp\Bigg(\frac{\frac{t^2}{16}\log^2\frac{x}{4\pi n^2} + 0.626}{x - 6.66}\Bigg) - 1,$$

the imaginary part of $\frac{1\pm y+ix}{2}$ being $\frac{x}{2}$. For $1 \leq n \leq N$ and x in the window of index N one has $\frac{x}{4\pi n^2} \leq \frac{x}{4\pi}$ and $\frac{x}{4\pi n^2} \geq \frac{X_N(t)}{4\pi N^2} = 1 - \frac{t}{16N^2}$, so

$$\Big|\log\frac{x}{4\pi n^2}\Big| \;\leq\; \max\Big(\log\frac{x}{4\pi}, -\log\Big(1 - \frac{t}{16N^2}\Big)\Big) \;=\; \log\frac{x}{4\pi},$$

the last step because $\frac{x}{4\pi}\big(1 - \frac{t}{16N^2}\big) \geq \frac{(N^2-t/16)^2}{N^2} \geq 1$ (indeed $N^2 - \frac{t}{16} \geq N$ as $N \geq N_*$). Hence the per-term exponent is at most $\varphi(x) = \big(\frac{t^2}{16}\log^2\frac{x}{4\pi} + 0.626\big)/(x - 6.66)$. Writing $L = \log\frac{x}{4\pi}$, which exceeds 2 for $x \geq 200$,

$$(x-6.66)^2\,\varphi'(x) = \frac{t^2}{8}\,L\,\frac{x-6.66}{x} - \Big(\frac{t^2}{16}L^2 + 0.626\Big) \;<\; \frac{t^2}{8}L - \frac{t^2}{16}L^2 - 0.626 = -\frac{t^2}{16}L(L-2) - 0.626 \;<\; 0,$$

so φ is decreasing, and on the window $\varphi(x) \leq \varphi(X_N(t)) = \Delta(N,t)$: the per-term factor is at most $e^{\Delta(N,t)} - 1$.

Assembly of E_{AB}. By (71)–(72) of [7] and Lemma 5.1 ($\operatorname{Re} s_* \geq \sigma_1$, $|\gamma| \leq G$, and $n^{y - \operatorname{Re} s_* - \operatorname{Re}\kappa} \leq n^{-\sigma_2}$),

$$e_B \leq \big(e^{\Delta(N,t)} - 1\big) \sum_{n \leq N} b_t(n)\, n^{-\sigma_1}, \qquad e_A \leq G\big(e^{\Delta(N,t)} - 1\big) \sum_{n \leq N} b_t(n)\, n^{-\sigma_2}.$$

Splitting each sum at M_{err} and bounding the range $M_{\mathrm{err}} < n \leq N$ by Lemma 5.2 (legitimate by (22)) gives $e_A + e_B \leq \mathrm{E}_{AB}(N,t,y)$.

Bound for $e_{C,0}$. By the corollary (13), $e_{C,0}$ is bounded by the product of $(\frac{x}{4\pi})^{-(1+y)/4}$, $\exp(-\frac{t}{16}\log^2\frac{x}{4\pi})$, $\exp\big(1.24(3^y{+}3^{-y})/(N{-}0.125)\big)$, and $\exp(\psi(x))$ with $\psi(x) = \big(3\sqrt{L^2 + \pi^2/4} + 10.50\big)/(x - 12)$. The first two factors are decreasing in x, the third does not depend on x, and ψ is decreasing: with $Q(L) = 3\sqrt{L^2 + \pi^2/4} + 10.50$ one has $\frac{d}{dx}Q = \frac{3L}{\sqrt{L^2+\pi^2/4}} \cdot \frac{1}{x} < \frac{3}{x}$, whence $(x-12)^2\,\psi'(x) < 3\,\frac{x-12}{x} - Q < 3 - 10.50 < 0$. Evaluating all four factors at the left edge $x = X_N(t)$, where $\log\frac{x}{4\pi} = L_N(t)$, gives $e_{C,0} \leq \mathrm{E}_C(N,t,y)$.

Decay in N. In E_{AB}: the factor $e^{\Delta(N,t)} - 1$ is nonincreasing because $\Delta(N,t) = \varphi(X_N(t))$ with φ decreasing and $X_N(t)$ increasing in N; the heads of S_1, S_2 and the factor G are nonincreasing by Lemma 5.6; and the caps $\mathrm{Cap}_t(M_{\mathrm{err}}, N; \sigma_i)$ are nonincreasing by Lemma 5.5 with $a = M_{\mathrm{err}}$, whose conditions (25) hold by hypothesis. In E_C: the factors $(N^2 - t/16)^{-(1+y)/4}$, $e^{-\frac{t}{16}L_N(t)^2}$ and $e^{1.24(3^y+3^{-y})/(N-0.125)}$ are visibly nonincreasing in N, and $e^{\psi(X_N(t))}$ is nonincreasing because ψ is decreasing and $X_N(t)$ increases. Products of nonnegative nonincreasing factors are nonincreasing. $\square$

5.6 The certified cutoff inequalities and the tail theorem

The computational input of the tail argument is the following.

Proposition 5.10 (Certified cutoff inequalities)**.** *The cap-validity conditions* (22) *and the cutoff conditions* (25)–(26) *hold, uniformly for $t \in I_t$ and $y \in [y_0, \sqrt{1-2t}]$, and*

$$\sup_{t\in I_t,\ y\in I_{\mathrm{box}}} D(N_*, t, y) < 0.999721, \qquad M_{\max} < 1.608290,$$

$$\sup_{t\in I_t,\ y\in I_{\mathrm{ext}}} \big(\mathrm{E}_{AB} + \mathrm{E}_C\big)(N_*, t, y) < 1.1672 \times 10^{-8},$$

with $\mathrm{E}_{AB}, \mathrm{E}_C$ the error majorants of Lemma 5.9. In particular

$$\frac{1-D}{M_{\max}} - (e_A + e_B + e_{C,0}) \ > \ 0.00017352 \ > \ 0.$$

Proof. The proof is computer-assisted; the verification program `prop510_proof.c`, included with this manuscript as supplementary material (directory `code/prop510`; derived from Gomila's source as documented in its header and reviewed line by line, check by check, as part of this audit), certifies every condition and every displayed constant of the statement. It reads no stored data; every decisive comparison is a strict ball inequality, which fails closed when enclosures overlap; and it was run at both 256 and 512 bits of precision with identical verdicts.

How the suprema are captured. The intervals I_t, I_{box} and I_{ext} enter the computation as single balls whose endpoints are the exact rational interval endpoints; consequently every derived quantity is an enclosure valid simultaneously for all (t, y) in the corresponding hull, and its directed upper (resp. lower) endpoint bounds the supremum (resp. infimum) over that hull. The frozen quantities $\widehat{\delta}$ and $\widehat{k}$ are extracted as point *upper* bounds over I_t, and the exponents σ_1, σ_2 as point *lower* bounds over the hulls; this is conservative throughout because every occurrence of σ_i is through $u^{-\sigma_i}$ or $E_{t,\sigma_i}(u)$ with $u \ge 1$, both nonincreasing in σ_i.

What is certified. (1) The cap-validity conditions (22), in the equivalent form $\sigma_i > \frac{t}{2}\log N_*$, on both hulls (labels `SC1`, `SC2` in the program's output). (2) The cutoff conditions (25) for exactly the left endpoints listed there: $a = \lfloor M/d \rfloor$ for each $d \in \mathcal{S}$ at σ_1 (`GN-TR`), $a = M$ at σ_2 (`GN-AB`), and $a = M_{\mathrm{err}}$ at both σ_1 and σ_2 (`GN-error`), each with the t-factor evaluated at $\min I_t$ and the σ-factor at its lower bound over I_{ext}. (3) The A-side slope condition (26) (`YM`). (4) The hypotheses of Lemmas 5.5–5.9: $\sigma_1, \sigma_2 > 1$ on I_{ext}, $N_*\big(1 - \frac{t}{16N_*^2}\big) > 1$, $X_{N_*}(t) > 200$, $\log(N_*^2 - \frac{t}{16}) > 2$, and the sign conditions making $\Delta(N, t)$ and the $e_{C,0}$ quotient decreasing. (5) The exact rational identities: the mollifier normalization $\sum_d |\lambda_d| = 9.918746$, $\lambda_1 = 1$, $t_0 + y_0^2/2 = \frac{893927}{5\cdot 10^6}$, and the hull containments $y_0 \in I_{\mathrm{box}}$, $\max I_{\mathrm{box}} \le \sqrt{1-2t}$ for all $t \in I_t$, and $\max I_{\mathrm{ext}} \ge \sqrt{1-2t_0}$, all verified in exact integer arithmetic. (6) The quantity D is assembled per (21) from an exact Dirichlet convolution $c_m(t)$ computed by direct double summation over (d, k), and the four displayed constants are certified as strict ball comparisons against exact rationals:

$$D < 0.999721, \qquad M_{\max} < 1.608290, \qquad \mathrm{E}_{AB} + \mathrm{E}_C < 1.1672 \times 10^{-8},$$

$$\frac{1-D}{M_{\max}} - (\mathrm{E}_{AB} + \mathrm{E}_C) \ > \ 0.00017352$$

(the last check was added in this review: Gomila's original program certified only positivity of this margin, together with two-sided "corridor" checks on each constituent, which are retained since they expose missing or duplicated terms). The certified enclosures at 512 bits include $D \leq 0.9997209094$, $M_{\max} \leq 1.6082891450$, $\mathrm{E}_{AB} + \mathrm{E}_C \leq 1.16716035 \times 10^{-8}$, and margin ≥ 0.0001735209. Both runs (256 and 512 bits) pass all 37 checks; the outputs are included as `code/prop510/runs/prop510_report_256.txt` and `prop510_report_512.txt`, and each run completes in under a second. $\square$

Theorem 5.11 (Tail theorem)**.** *For every $t \in I_t$, $y \in [y_0, \sqrt{1-2t}]$ and $x \geq X_{N_*}(t)$,*

$$|f_t(x+iy)| \;>\; e_A + e_B + e_{C,0},$$

and consequently $H_t(x+iy) \neq 0$ on this region.

Proof. Let N be the window index of x, so $N \geq N_*$. By Lemma 5.3, Lemma 5.8 and Proposition 5.10,

$$|f_t(x+iy)| \;\geq\; \frac{1-D(N,t,y)}{M_{\max}} \;\geq\; \frac{1-0.999721}{M_{\max}} \;>\; 0.0001735 \;>\; 1.1672\times 10^{-8} \;\geq\; e_A+e_B+e_{C,0},$$

using in the second step both clauses of Lemma 5.8 with Proposition 5.10 ($D < 0.999721$ and $M_{\max} < 1.608290$), and in the last step Lemma 5.9 together with Proposition 5.10 and $[y_0, \sqrt{1-2t}] \subseteq I_{\text{ext}}$. The point (x, y, t) lies in the region (7), so [7, Corollary 1.4] gives $H_t(x+iy) \neq 0$. $\square$

6 Hypothesis (iii): the barrier certificate

6.1 Statement, and reduction to a closed rectangle

Throughout this section let

$$R = [X,\, X+1] + i\Big[\tfrac{1809}{10000},\, 1\Big].$$

The goal of the section is the following theorem, whose proof is assembled in Section 6.5.

Theorem 6.1 (Barrier theorem)**.** *$H_t(z) \neq 0$ for all $z \in R$ and $0 \leq t \leq t_0$.*

Theorem 6.1 implies hypothesis (iii) of Theorem 2.1: the curved barrier region of that hypothesis satisfies, horizontally, $X \leq x \leq X + \sqrt{1-y_0^2} \leq X+1$; and vertically, $y \geq \sqrt{y_0^2 + 2(t_0 - t)} \geq y_0 > \frac{1809}{10000}$ (indeed $y_0^2 - \big(\frac{1809}{10000}\big)^2 = \frac{234599}{10^8} > 0$) and $y \leq \sqrt{1-2t} \leq 1$; hence it is contained in $R \times [0, t_0]$.

6.2 Uniform effective error on the box, including $t = 0$

Proposition 6.2 (Uniform error bound on the barrier box)**.** *For all $x \in [X, X+1]$ and $0 \leq t \leq t_0$, the window index satisfies $N = \big\lfloor \sqrt{\frac{x}{4\pi} + \frac{t}{16}} \big\rfloor = 690988$ and $N^2 < x/(4\pi)$. Moreover, for all $z \in R$ and $0 < t \leq t_0$, the error terms of Theorem 1.3, bounded with the constants of Section 2.3, satisfy*

$$e_A + e_B + e_{C,0} \;<\; 0.000356523011600040 \;<\; 0.00125. \tag{27}$$

Proof. The proof is computer-assisted; the verification program `prop62_proof.c`, included with this manuscript as supplementary material (directory `code/prop62`; derived from Gomila's source as documented in its header and reviewed line by line as part of this audit), encloses every quantity below in 256-bit ball arithmetic starting from the exact inputs X, $t_0 = \frac{129}{800}$, $y_{\min} = \frac{1809}{10000}$ and $N = 690988$; it reads no stored data, and every acceptance condition is a strict ball inequality, which fails closed.

Window-index constancy. The quantity $q(x,t) = \frac{x}{4\pi} + \frac{t}{16}$ is strictly increasing in each of x and t, and $N = \lfloor \sqrt{q} \rfloor$ is nondecreasing in q; so it suffices to evaluate N at the two extreme corners $(X, 0)$ and $(X+1, t_0)$ of the box. At each corner the program produces an enclosure of $\lfloor \sqrt{q} \rfloor$ that collapses to an exact integer (a straddling enclosure would fail closed) and checks that both integers equal 690988. It further certifies $N^2 < X/(4\pi) \le x/(4\pi)$, and the equivalent inequality $2 \log N < L_{\min} := \log \frac{X}{4\pi}$; write also $L_{\max} := \log \frac{X+1}{4\pi}$.

Error bound. With N constant, for $0 < t \le t_0$ the error terms are bounded by equation (23) of [7] and by (13); the program encloses the following majorants, each uniform on the box because every t- and y-dependent factor is evaluated at its worst endpoint: (i) by (11), using $(\,\cdot\,)_+ \le 1 + 8/X^2$ and $y \ge y_{\min}$, $\operatorname{Re} s_* \ge a + \frac{t}{4} L_{\min} - \delta$ with $a = \frac{1+y_{\min}}{2}$ and $\delta = \frac{t_0(1+8/X^2)}{2X^2}$; since $\log n \le \log N < \frac{1}{2} L_{\min}$, this gives $b_t(n)\, n^{-\operatorname{Re} s_*} \le n^{-a_{\text{eff}}}$ with $a_{\text{eff}} = a - \delta$, uniformly in $t \in [0, t_0]$ (the t-dependent parts cancel against $\frac{t}{4} \log^2 n$; certified: $0 < a_{\text{eff}} < 1$); (ii) hence, by comparison with $\int_1^N u^{-a_{\text{eff}}}\, du$, $\sum_{n=1}^N b_t(n)\, n^{-\operatorname{Re} s_*} \le 1 + \frac{N^{1-a_{\text{eff}}}-1}{1-a_{\text{eff}}}$; (iii) since $n^2 \le N^2 < x/(4\pi)$, the bounds (10) and (12) give $|\gamma| n^y \le e^{0.02y} \big(\frac{4\pi n^2}{x}\big)^{y/2} \le e^{0.02}$ and $N^{|\kappa|} \le \exp\big(\frac{t_0 \log N}{2(X-6)}\big)$, so the prefactor of (23) is at most $1 + e^{0.02} \exp\big(\frac{t_0 \log N}{2(X-6)}\big)$; (iv) since $0 < \log \frac{x}{4\pi n^2} \le L_{\max}$, the last factor of (23) is at most $\exp\big(\frac{(t_0^2/16) L_{\max}^2 + 0.626}{X - 6.66}\big) - 1$; (v) in (13), the prefactor is maximized at $(x, y) = (X, y_{\min})$, the factor $e^{-\frac{t}{16} \log^2 \frac{x}{4\pi}}$ is at most 1, $3^y + 3^{-y} \le 3 + \frac{1}{3} = \frac{10}{3}$ (increasing in y, and $y \le 1$), and the final quotient is bounded by maximizing its numerator ($L \le L_{\max}$) and minimizing its denominator ($x \ge X$) separately. The program multiplies and sums these enclosures — yielding $e_A + e_B \le 3.64 \times 10^{-10}$ and $e_{C,0} \le 0.000356522648$ — and certifies the two strict inequalities of (27), the sharp constant being compared as the exact rational $356523011600040 \times 10^{-18}$. The run completes in under a second; its output is included as `code/prop62/runs/prop62_report.txt`. □

The constancy of N makes f_t a single analytic expression on R. The round number $0.00125 = \frac{1}{800}$ is the *approximation allowance* used by the prism criterion below; (27) says it is safely conservative.

Theorem 1.3 is stated in [7] for $t > 0$; the following lemma extends its conclusion to the closed endpoint $t = 0$ on the compact box.

Lemma 6.3 (Extension to $t = 0$)**.** *For all $z \in R$ and all $0 \le t \le t_0$ (including $t = 0$), $B_t(z) \ne 0$ and*

$$\big| g_t(z) - f_t(z) \big| \;\le\; 0.00125,$$

where $g_t = H_t / B_t$ and f_t is the sum (9) with the constant window index $N = 690988$.

Proof. For $0 < t \le t_0$ this is Theorem 1.3 together with Proposition 6.2. For the endpoint, fix $z \in R$ and argue by a limit $t_j \downarrow 0$: (1) by dominated convergence in (1) (majorant $e^{t_0 u^2 + u} |\Phi(u)|$, integrable by the super-exponential decay of Φ), $H_t(z)$ is jointly continuous down to $t = 0$ on R; (2) $B_t^{\pm 1}$ is continuous and $B_t \ne 0$ on $R \times [0, t_0]$, because the argument $\frac{1+y-ix}{2}$ of M_0 stays away from the branch cut (Section 2.2) and the deforming exponential never vanishes; (3) since the window index is constant, f_t is a fixed finite sum of functions continuous in t; (4) the error majorant is continuous in t and bounded by 0.00125 on $(0, t_0]$ by (27). Passing to the limit in $|g_{t_j} - f_{t_j}| \le 0.00125$ gives the claim at $t = 0$. $\square$

6.3 Uniform derivative bounds on the box

The moving-mesh argument below needs uniform bounds for $|\partial_z f_t|$ and $|\partial_t f_t|$ on R. The pointwise input is the following result; recall that by Proposition 6.2 the window index is constantly $N = 690988$ on the box, so that f_t is the fixed finite sum (9) with this N, jointly smooth in (z, t), and the lemma applies to it verbatim.

Lemma 6.4 ([7, Lemma 8.4])**.** *In the region (7), and away from the jump discontinuities of N,*

$$\Big|\frac{\partial f_t}{\partial z}\Big| \le \sum_{n=1}^{N} \frac{b_t(n)}{n^{\operatorname{Re} s_*}}\Big(\frac{\log n}{2} + \frac{t\log n}{4(x-6)}\Big) + |\gamma| N^{|\kappa|} \sum_{n=1}^{N} \frac{b_t(n)\, n^y}{n^{\operatorname{Re} s_*}}\Big(\frac{t\log n}{4(x-6)} + \Big(\log\frac{|1+y+ix|}{4\pi} + \pi + \frac{3}{x}\Big)\Big(\frac{1}{2} + \frac{t}{4(x-6)}\Big)\Big),$$

and

$$\Big|\frac{\partial f_t}{\partial t}\Big| \le \sum_{n=1}^{N} \frac{b_t(n)}{n^{\operatorname{Re} s_*}}\Big(\frac{1}{4}\log n \log\frac{x}{4\pi n} + \frac{\pi}{8}\log n + \frac{2\log n}{x-6}\Big) + |\gamma| N^{|\kappa|} \sum_{n=1}^{N} \frac{b_t(n)\, n^y}{n^{\operatorname{Re} s_*}}\Big(\frac{1}{4}\log n\log\frac{x}{4\pi n} + \frac{\pi}{8}\log n + \frac{2\log n}{x-6} + \frac{1}{4}\Big(\frac{\pi}{2} + \frac{8}{x-6}\Big)\Big(\log\frac{x}{4\pi} + \frac{8}{x-6}\Big)\Big).$$

To convert these pointwise bounds into uniform ones we introduce the following frozen quantities. Write $y_{\min} = \frac{1809}{10000}$ (the bottom of R), $q(x) = \frac{x}{4\pi}$, and, as in (11),

$$h(x,y) = 1 - 3y + \frac{4y(1+y)}{x^2}, \qquad c(x,y) = \frac{1}{4}\log q(x) - \frac{h(x,y)_+}{2x^2}, \qquad c_0 = c(X, y_{\min}),$$

so that (11) reads $\operatorname{Re} s_* \ge \frac{1+y}{2} + t\, c(x,y)$. Define further

$$\begin{aligned}
a(t) &= e^{0.02 y_{\min}}\, q(X)^{-y_{\min}/2}\, N^{t\, y_{\min}/(2(X-6))},\\
P_t(u) &= u^{-\frac{1+y_{\min}}{2} + t(\frac{1}{4}\log u - c_0)},\\
\theta(t) &= \frac{1}{2} + \frac{t}{4(X-6)},\\
\Lambda_z(t) &= \Big(\log\frac{|2+i(X+1)|}{4\pi} + \pi + \frac{3}{X}\Big)\theta(t),\\
\Lambda_t &= \frac{1}{4}\Big(\frac{\pi}{2} + \frac{8}{X-6}\Big)\Big(\log\frac{X+1}{4\pi} + \frac{8}{X-6}\Big),\\
C_2 &= \frac{1}{4}\log\frac{X+1}{4\pi} + \frac{\pi}{8} + \frac{2}{X-6},
\end{aligned}$$

and the two summand majorants

$$F_z(u;t) = P_t(u)\Big[\theta(t)\log u + a(t)\,u^{y_{\min}}\Big(\frac{t\log u}{4(X-6)} + \Lambda_z(t)\Big)\Big],$$
$$F_t(u;t) = P_t(u)\Big[\big(1 + a(t)\,u^{y_{\min}}\big)\log u\,\Big(C_2 - \frac{\log u}{4}\Big) + a(t)\,u^{y_{\min}}\Lambda_t\Big],$$

and finally

$$D_z(t) = \sum_{n=1}^{16} F_z(n;t) + \int_{16}^{N} F_z(u;t)\,du, \qquad D_t(t) = \sum_{n=1}^{16} F_t(n;t) + \int_{16}^{N} F_t(u;t)\,du.$$

Two tight inequalities at the exact barrier parameters are needed below. The first, $N^2 < q(X)$, is part of Proposition 6.2. The second we record as a standalone statement.

Proposition 6.5 (Certified tail-exponent inequality)**.** *At $X = 6000000185827$, $N = 690988$, $y_{\min} = \frac{1809}{10000}$,*

$$\frac{1}{4}\log\frac{q(X)}{N^2} \;>\; \frac{h(X, y_{\min})_+}{2X^2}.$$

Proof. The proof is computer-assisted; the verification program `prop65_proof.c`, included with this manuscript as supplementary material (directory `code/prop65`; written for this review as a standalone extraction of the corresponding runtime check of Gomila's barrier program, as documented in its header), encloses both sides in 256-bit ball arithmetic from the exact rational parameters and certifies the displayed strict inequality as a strict ball comparison, which fails closed. The certified enclosures are

$$\tfrac{1}{4}\log\tfrac{q(X)}{N^2} = 2.2405813008\ldots\times 10^{-7} \pm 10^{-31}, \qquad \frac{h(X, y_{\min})_+}{2X^2} = 6.3513884954\ldots\times 10^{-27} \pm 10^{-51};$$

the inequality is tight only in the sense that its left side is small — it is positive precisely because $N^2 < q(X)$, an inequality of relative size 3×10^{-7} — while the right side is smaller by twenty orders of magnitude. The run completes in milliseconds; its output is included as `code/prop65/runs/prop65_report.txt`. The same inequality is also re-checked at run time by the barrier program of Proposition 6.12 (harmlessly, like the conditions (C2), (C4) and (C7) of Proposition 6.6). □

The construction consumes the following list of strict inequalities at the exact barrier parameters.

Proposition 6.6 (Derivative-bound conditions)**.** *At $X = 6000000185827$, $N = 690988$, $y_{\min} = \frac{1809}{10000}$, $t_0 = \frac{129}{800}$:*

(C1) $X^2 > 8$ *and* $\frac{1}{4(X+1)} - \frac{2}{X^3} > 0$*;*

(C2) $0.02 - \frac{1}{2}\log q(X) + \frac{t_0 \log N}{2(X-6)} + \frac{1}{2}\log N < 0$*;*

(C3) $2\log N < \log q(X)$*;*

(C4) $\frac{12}{X^2} < 3$*;*

(C5) $c_0 > \frac{1}{2}\log N$;

(C6) $\frac{1-y_{\min}}{2} > \frac{1}{\log 16}$;

(C7) $C_2 > \frac{1}{4}\log N$.

Proof. (C1), (C4) and (C6) are immediate by inspection ($X \approx 6\times 10^{12}$; $\frac{1}{\log 16} = 0.3607\ldots < 0.40955 = \frac{1-y_{\min}}{2}$). For (C2), rewrite the left side as $0.02 + \frac{1}{2}\log\frac{4\pi N}{X} + \frac{t_0\log N}{2(X-6)}$: here $\frac{4\pi N}{X} < \frac{8.8\times 10^6}{6\times 10^{12}} < 2\times 10^{-6} < e^{-13}$, so the middle term is less than $-\frac{13}{2}$, while the last term is less than 10^{-12}; hence the left side is less than $0.02 - 6.5 + 10^{-12} < 0$. (C7) follows from (C3): $C_2 - \frac{1}{4}\log N \geq \frac{1}{4}\log q(X) - \frac{1}{4}\log N > \frac{1}{4}\log N > 0$.

The remaining two conditions are exactly the statements established earlier: (C3) is the inequality $N^2 < q(X)$ of Proposition 6.2, and (C5) is Proposition 6.5, since $c_0 - \frac{1}{2}\log N = \frac{1}{4}\log\frac{q(X)}{N^2} - \frac{h(X,y_{\min})_+}{2X^2}$. (The barrier program also re-checks (C2), (C4) and (C7) at run time; by the above, those checks are mathematically redundant, though harmless.) $\square$

Lemma 6.7 (Pointwise domination on the box)**.** *Assume (C1), (C2), (C4). Then for all $x \in [X, X+1]$, $y \in [y_{\min}, 1]$, $0 \leq t \leq t_0$ and $1 \leq n \leq N$, the n-th summand of the first (resp. second) bound of Lemma 6.4 is at most $F_z(n;t)$ (resp. $F_t(n;t)$). Consequently, for $0 \leq t \leq t_0$,*

$$\sup_{z\in R}\Big|\frac{\partial f_t}{\partial z}(z)\Big| \leq \sum_{n=1}^{N} F_z(n;t), \qquad \sup_{z\in R}\Big|\frac{\partial f_t}{\partial t}(z)\Big| \leq \sum_{n=1}^{N} F_t(n;t).$$

Proof. Step 1: c is minimized at $(X, y_{\min})$. In y: $\partial_y h = -3 + \frac{4(1+2y)}{x^2} \leq -3 + \frac{12}{X^2} < 0$ by (C4), so h, hence h_+, is nonincreasing in y and c is nondecreasing in y. In x: where $h > 0$,

$$\Big|\partial_x \frac{h}{2x^2}\Big| = \Big|-\frac{4y(1+y)}{x^5} - \frac{h}{x^3}\Big| \leq \frac{8}{x^5} + \frac{h_+}{x^3} \leq \frac{2}{x^3},$$

using $4y(1+y) \leq 8$, $8/x^2 \leq 1$ and $h_+ \leq 1$ (both from $X^2 > 8$; for $y > 0$, $h \leq 1$ reduces to $4(1+y) \leq 3x^2$, and at $y = 0$, $h = 1$). Where $h < 0$ the positive part is locally constant, and $h_+/(2x^2)$ is locally Lipschitz across the kink with the same bound. Hence for $x \in [X, X+1]$, $\partial_x c \geq \frac{1}{4x} - \frac{2}{X^3} \geq \frac{1}{4(X+1)} - \frac{2}{X^3} > 0$ almost everywhere by (C1), so c is increasing in x. Therefore $c(x,y) \geq c(X, y_{\min}) = c_0$ on the box.

Step 2: the B-side. By (11) and Step 1, $\operatorname{Re} s_* \geq \frac{1+y}{2} + t\,c(x,y) \geq \frac{1+y_{\min}}{2} + t\,c_0$, so

$$b_t(n)\,n^{-\operatorname{Re} s_*} \leq n^{\frac{t}{4}\log n - \frac{1+y_{\min}}{2} - t\,c_0} = P_t(n),$$

while the B-side brackets obey $\log n\big(\frac{1}{2} + \frac{t}{4(x-6)}\big) \leq \theta(t)\log n$ and

$$\log n\Big(\frac{1}{4}\log\frac{x}{4\pi n} + \frac{\pi}{8} + \frac{2}{x-6}\Big) = \log n\Big(\frac{1}{4}\log q(x) - \frac{\log n}{4} + \frac{\pi}{8} + \frac{2}{x-6}\Big) \leq \log n\Big(C_2 - \frac{\log n}{4}\Big),$$

since q is increasing and $\frac{1}{x-6}$ decreasing on $[X, X+1]$.

Step 3: the A-core. By (10)–(12), $|\gamma| N^{|\kappa|} n^y\, b_t(n)\, n^{-\operatorname{Re} s_*} \leq b_t(n)\,\mathcal{A}(x,y,t;n)$ with

$$\mathcal{A}(x,y,t;n) = e^{0.02y} q(x)^{-y/2} N^{\frac{ty}{2(x-6)}}\, n^{y - \frac{1+y}{2} - t\,c(x,y)}.$$

Using $\partial_y c \geq 0$ (Step 1) and $\log n \geq 0$,

$$\partial_y \log \mathcal{A} = 0.02 - \frac{1}{2}\log q(x) + \frac{t\log N}{2(x-6)} + \frac{\log n}{2} - t\,(\partial_y c)\log n \leq 0.02 - \frac{1}{2}\log q(X) + \frac{t_0 \log N}{2(X-6)} + \frac{\log N}{2},$$

which is negative by (C2); so $\mathcal{A}$ is maximized at $y = y_{\min}$. At $y = y_{\min}$, freezing x conservatively ($q(x)^{-y_{\min}/2} \leq q(X)^{-y_{\min}/2}$, $N^{\frac{ty_{\min}}{2(x-6)}} \leq N^{\frac{ty_{\min}}{2(X-6)}}$, $c(x, y_{\min}) \geq c_0$) gives $b_t(n)\,\mathcal{A} \leq a(t)\, n^{y_{\min}} P_t(n)$.

Step 4: the A-side brackets. On $[X, X+1] \times [y_{\min}, 1]$, we have

$$\frac{t\log n}{4(x-6)} \leq \frac{t\log n}{4(X-6)},$$

$$\Big(\log\frac{|1+y+ix|}{4\pi} + \pi + \frac{3}{x}\Big)\Big(\frac{1}{2} + \frac{t}{4(x-6)}\Big) \leq \Big(\log\frac{|2+i(X+1)|}{4\pi} + \pi + \frac{3}{X}\Big)\theta(t) = \Lambda_z(t),$$

since $|1+y+ix| \leq |2+i(X+1)|$ (both coordinates increase) and $\frac{3}{x} \leq \frac{3}{X}$; and

$$\frac{1}{4}\Big(\frac{\pi}{2} + \frac{8}{x-6}\Big)\Big(\log q(x) + \frac{8}{x-6}\Big) \leq \Lambda_t,$$

each factor being bounded by its value at the appropriate endpoint.

Combining Steps 2–4 termwise bounds the two summands of Lemma 6.4 by $F_z(n;t)$ and $F_t(n;t)$. For $0 < t \leq t_0$ the displayed conclusions are then Lemma 6.4; at $t = 0$ they follow by continuity, as both sides of each inequality are continuous in t on the compact box (f_t being the fixed finite sum). □

Lemma 6.8 (Head-and-tail evaluation)**.** *Assume (C5), (C6), (C7). Then for each fixed $t \in [0, t_0]$ the functions $u \mapsto F_z(u;t)$ and $u \mapsto F_t(u;t)$ are positive and nonincreasing on $[16, N]$, and consequently*

$$\sum_{n=1}^{N} F_z(n;t) \;\leq\; D_z(t), \qquad \sum_{n=1}^{N} F_t(n;t) \;\leq\; D_t(t).$$

Proof. Write $r = \log u \in [\log 16, \log N]$. The logarithmic derivative of P_t with respect to r is $p(r) = -\frac{1+y_{\min}}{2} + t\big(\frac{r}{2} - c_0\big) \leq -\frac{1+y_{\min}}{2}$, since $t \geq 0$ and $\frac{r}{2} \leq \frac{1}{2}\log N \leq c_0$ by (C5). Each of F_z, F_t is a nonnegative linear combination (with u-independent coefficients $\theta(t)$, $\frac{t}{4(X-6)}$, $a(t)$, $\Lambda_z(t)$, Λ_t, all ≥ 0) of the six component functions

$$P_t(u)\,r, \quad P_t(u)\,u^{y_{\min}} r, \quad P_t(u)\,u^{y_{\min}}, \quad P_t(u)\,B(r), \quad P_t(u)\,u^{y_{\min}} B(r),$$

with $B(r) = r\,(C_2 - \frac{r}{4})$, which is positive on $[\log 16, \log N]$ by (C7). Their logarithmic r-derivatives are, respectively, at most

$$p + \frac{1}{r}, \quad p + y_{\min} + \frac{1}{r}, \quad p + y_{\min}, \quad p + \frac{B'}{B}, \quad p + y_{\min} + \frac{B'}{B},$$

and $\frac{B'}{B} = \frac{1}{r} - \frac{1}{4(C_2 - r/4)} \leq \frac{1}{r}$. Hence every component has logarithmic derivative at most

$$p + y_{\min} + \frac{1}{\log 16} \leq -\frac{1+y_{\min}}{2} + y_{\min} + \frac{1}{\log 16} = -\frac{1-y_{\min}}{2} + \frac{1}{\log 16} \;<\; 0$$

by (C6), so each component, and therefore each of F_z, F_t, is positive and decreasing on $[16, N]$. Finally, for decreasing positive F, $\sum_{n=17}^{N} F(n) \leq \int_{16}^{N} F(u)\,du$ (compare each term with the integral over $[n-1, n]$), which gives the displayed bounds. □

Lemma 6.9 (Uniform derivative bounds)**.** *Assume (C1)–(C7). Then for every $t \in [0, t_0]$,*

$$\sup_{z\in R}\Big|\frac{\partial f_t}{\partial z}(z)\Big| \;\le\; D_z(t), \qquad \sup_{z\in R}\Big|\frac{\partial f_t}{\partial t}(z)\Big| \;\le\; D_t(t),$$

and hence, for every closed interval $[t_i, t_{i+1}] \subseteq [0, t_0]$, the relation

$$\sup_{z\in R,\ \tau\in[t_i,t_{i+1}]} |\partial_\tau f_\tau(z)| \le \sup_{\tau\in[t_i,t_{i+1}]} D_t(\tau)$$

holds.

Proof. Combine Lemmas 6.7 and 6.8. □

Remark 6.10. The barrier program evaluates $D_z(t)$ and $D_t(t)$ with t an interval enclosure — for the prism bound, an enclosure of the whole interval $[t_i, t_{i+1}]$, which by inclusion isotonicity encloses $\sup_\tau D_t(\tau)$ — and computes the integrals $\int_{16}^N F(u)\,du$ by Arb's rigorous Gauss–Legendre integrator [4]. That algorithm requires the integrand callback to supply enclosures that are holomorphic on the complex balls it queries; the integrands above extend holomorphically in u away from the branch cut of the logarithm, and the program uses the analytic logarithm and power operations, which return non-finite enclosures (failing the run) if a queried ball touches the cut. Two interfaces of the original Polymath-era barrier code are thereby repaired: the discrete sums are no longer bounded by the unjustified expression $F(1) + \int_1^N F$ (the summand is not monotone near $n = 1$; Lemma 6.8 certifies monotonicity only from $n = 16$ on), and the callbacks no longer project the integration variable to its real part, which would void the integrator's error analysis.

6.4 The prism criterion

The time interval $[0, t_0]$ will be covered by finitely many closed "prisms" $R \times [t_i, t_{i+1}]$. The following lemma reduces nonvanishing of H on a prism to finitely many verifiable conditions at its initial time t_i.

Lemma 6.11 (Prism criterion)**.** *Let $0 \le t_i < t_{i+1} \le t_0$, and let $z_1, \dots, z_m$ (cyclically ordered, $z_{m+1} = z_1$) be a mesh on ∂R with consecutive spacing at most h. Let $D_z \ge \sup_{z\in R} |\partial_z f_{t_i}(z)|$ and $D_t \ge \sup_{z\in R,\ \tau\in[t_i,t_{i+1}]} |\partial_\tau f_\tau(z)|$ (as supplied by Lemma 6.9), and suppose:*

(a) *$\big|f_{t_i}(z_j)\big| \ge M_i$ for all j, where*

$$M_i \;>\; \frac{D_z h}{2} + D_t\,(t_{i+1} - t_i) + 0.00125\,; \tag{28}$$

(b) *each argument increment $\arg\big(f_{t_i}(z_{j+1})/f_{t_i}(z_j)\big)$ lies strictly in $(-\pi, \pi)$, and the total winding of the polygon $\big(f_{t_i}(z_j)\big)_j$ about 0 is 0.*

Then $H_\tau(z) \ne 0$ for all $z \in R$ and $\tau \in [t_i, t_{i+1}]$.

Proof. On each half of a mesh subedge, both the true image curve $F(s) = f_{t_i}(\text{subedge}(s))$ and its endpoint chord lie in the closed disk of radius $D_z h/2$ about the nearer endpoint value (this is the factor $\frac{1}{2}$ in (28)); each disk is convex, so the straight-line homotopy from the true boundary curve to the mesh polygon stays inside these disks. Moving to time $\tau \in [t_i, t_{i+1}]$ displaces values by at most $D_t(\tau - t_i)$, and passing from f_τ to $g_\tau = H_\tau/B_\tau$ by at most 0.00125 (Lemma 6.3). Condition (28) says every resulting disk, centered at a mesh value of modulus $\geq M_i$, excludes the origin; hence the entire three-stage homotopy (polygon $\to$ true f_{t_i}-boundary image $\to$ $f_\tau(\partial R) \to g_\tau(\partial R)$) avoids zero. In particular each chord avoids the origin, so its argument variation is the principal argument of the ratio of its endpoint values and the winding number of the polygon about the origin is a well-defined integer; by (b) it is 0, and the zero-avoiding homotopy gives winding number 0 for $g_\tau(\partial R)$ as well, for every $\tau \in [t_i, t_{i+1}]$. Since $B_\tau \neq 0$ and B_τ is holomorphic on a neighborhood of R (Lemma 6.3 and Section 2.2), g_τ is holomorphic there and has the same zeros as H_τ, and g_τ is zero-free on ∂R; the argument principle then gives $H_\tau \neq 0$ on R. □

6.5 The certified prism decomposition

The computational input of the barrier argument is the following.

Proposition 6.12 (Certified prism decomposition)**.** *There exist times $0 = \tau_0 < \tau_1 < \cdots < \tau_{883}$ with $\tau_{883} \geq t_0$ such that for each $0 \leq i < 883$, the hypotheses (a)–(b) of Lemma 6.11 hold for the prism $R \times [\tau_i, \min(\tau_{i+1}, t_0)]$, with certified margin in (28) at least 0.5198498946 in every prism.*

Proof. The proof is computer-assisted; the verification program `prop612_proof.c`, included with this manuscript as supplementary material (directory `code/prop612`; assembled for this review from three components of Gomila's barrier pipeline, each descended from the Polymath codebase, as documented in its header and reviewed line by line as part of this audit), is fully self-contained: it reads no stored data, and every decisive comparison is a strict Arb ball comparison, which fails closed. It runs in three phases.

(1) Domain hypotheses and truncation. The program first certifies, at 192-bit precision: the uniform expansion-parameter bound $|w| \leq 0.6480 < 0.66$ on the whole slab (via the shift bounds $|\operatorname{Re}(\log n_0 - \alpha)| < 0.694$ and $|\operatorname{Im}(\log n_0 - \alpha)| < 0.7855$ for both α-arguments used below, where $n_0 = N/2$); the bound $0.0882 < 0.089$ for the modulus of the γ-type factor, both for its exact expression (on a 256×128 subdivision of the box) and for the implemented majorant; and, by a rigorous direct sum over $n \leq N$ with geometric tail factors, the complete two-variable 62-term Taylor truncation bound $1.9543 \times 10^{-22} < 10^{-20}$ — the allowance that is attached as an error ball to every boundary mesh value in phase (3).

(2) Coefficient generation. The 62×62 complex matrix

$$\text{finalmat}[e][k] = \sum_{n=1}^{N} n^{\frac{i(X+\frac{1}{2})-1}{2}} \frac{\left(\log \frac{n}{n_0}\right)^e}{e!} \frac{\left(\frac{1}{4}\log^2 \frac{n}{n_0}\right)^k}{k!}$$

is computed in memory by direct summation, each entry a genuine ball enclosure. The reconstruction identity — verified symbolically in the course of this review — shows that the polynomial evaluation of phase (3) applied to this matrix reproduces, exactly up to the certified truncation of phase (1), the two sums of (9) at the evaluation point, with the heat factor $b_t(n)$ emerging from the algebra of the two Taylor directions. (In Gomila's pipeline this matrix was read from a stored 20-digit file and independently regenerated for comparison; computing it in-program from the defining series removes the stored file entirely and settles the provenance of the coefficients.)

(3) The adaptive prism loop. Before the loop the program re-certifies the conditions (C1)–(C7) of Proposition 6.6 at run time. Then, starting at $\tau_0 = 0$: at each prism start τ_i it evaluates the derivative bounds $D_z(\tau_i)$ and (for the candidate prism, with the time parameter interval-valued over the entire closed prism) D_t, by the head-plus-integral quadratures of Section 6.3, rounding both up to exact integers; meshes ∂R with $4\lceil D_z \rceil - 4$ points (spacing at most $h = 1/(\lceil D_z \rceil - 1)$); requires every mesh enclosure of f_{τ_i} to exclude zero, every argument increment to lie strictly in $(-\pi, \pi)$, and the per-rectangle winding enclosure to lie strictly in $(-\frac{1}{4}, \frac{1}{4})$, certifying the integer winding number 0; chooses τ_{i+1} adaptively at half the certified slack of (28), as an exact dyadic number; and finally re-checks the complete gate $\min_j |f_{\tau_i}(z_j)| > \frac{D_z h}{2} + D_t(\tau_{i+1} - \tau_i) + 0.00125$ as a strict ball inequality. The loop ends only when the prisms cover $[0, t_0]$ exactly (adjacent seams are identical exact dyadic numbers, the first time is exactly 0, the final endpoint encloses $\frac{129}{800}$), and the aggregate winding enclosure must lie strictly in $(-\frac{1}{4}, \frac{1}{4})$. (The final prism may extend slightly beyond t_0; restricting it to its intersection with $[0, t_0]$ only shrinks the displacement terms in (28), so the certificate for the full final prism covers the truncated one, as required by the statement.)

The fresh run certifies 883 prisms — the same count as Gomila's sealed transcripts — with aggregate winding enclosure $[\pm 1.18 \times 10^{-14}]$ and worst certified margin $0.519849894613872543 65 \pm 2 \times 10^{-20}$, certified against the exact rational threshold $5198498946 \times 10^{-10}$ (a check added in this review; Gomila's original certified only positivity of each margin). The run completes in under eight minutes; its output is included as `code/prop612/runs/prop612_report.txt`. □

Proof of Theorem 6.1. By Proposition 6.12 and Lemma 6.11, $H_\tau(z) \neq 0$ for $z \in R$ and $\tau \in [\tau_i, \min(\tau_{i+1}, t_0)]$, for each i. The union of these intervals is $[0, t_0]$. □

7 Assembly: proof of Theorem 1.1

Proof of Theorem 1.1 (candidate argument). The parameters (4) satisfy $0 < t_0 < \frac{1}{2}$ and $0 < y_0^2 < 1 - 2t_0$. Hypothesis (i) of Theorem 2.1 holds by Section 3, since $X/2 < T_{\mathrm{PT}}$. Hypothesis (ii) holds by combining Proposition 4.11 (nonvanishing of H_{t_0} for $x_* \le x < x_{3840001}$, $y_0 \le y \le \sqrt{1 - 2t_0}$) with Theorem 5.11 applied at $t = t_0 \in I_t$ (nonvanishing for $x \ge x_{3840000} = X_{N_*}(t_0)$): the two ranges overlap on the full window $W_{3840000}$, so together they cover all $x \ge x_*$. Hypothesis (iii) holds by Theorem 6.1, since the curved barrier region is contained in $R \times [0, t_0]$ (Section 6). Theorem 2.1 then gives

$$\Lambda \ \le\ t_0 + \frac{y_0^2}{2} \ = \ \frac{893927}{5000000} \ = \ 0.1787854. \qquad \square$$

8 Summary of the verification

This section records the status of the verification.

Published inputs. The statements cited from [7] and [6] have been checked verbatim against the arXiv source files and are applied on domains where their hypotheses hold, the one caveat being Remark 2.2 (the constant 10.44 of [7, eq. (24)] is not used; the conservative 10.50 corollary (13) is derived and used instead).

Analytic arguments. Every analytic lemma carries a complete proof verified in the course of the review: the proofs of Lemmas 5.1, 5.3, 6.3 and 6.11 were verified in detail, and the formerly outlined proofs of Lemma 5.8, Lemma 6.9 and Proposition 4.4 were completed in full, through

Lemmas 5.5–5.7 and 5.9, Lemmas 6.7 and 6.8 with Proposition 6.6, and Lemmas 4.5–4.8 with Proposition 4.9, respectively. No analytic items remain.

Computer-assisted components. Each of the propositions whose proofs are computer-assisted — 4.3 (window sweep), 4.9 (Dini cells), 4.10 (finite-region error bound), 5.10 (tail cut-off inequalities), 6.2 (barrier-box error bound), 6.5 (tail-exponent inequality) and 6.12 (prism decomposition) — has been independently reproduced by a standalone program included as supplementary material (directory `code/`), derived from Gomila's sources as documented in each program's header, reviewed line by line against the corresponding statement in this paper, compiled against an independently built FLINT/Arb toolchain, reading no stored data, and with every displayed constant of the statements certified by an explicit strict comparison against an exact rational. The proofs given with the propositions record the protocols and the fresh-run results. In particular the stored certificate files, transcripts and coefficient data of Gomila's repository are nowhere consumed: every number entering Theorem 1.1 is either proved analytically in this paper or recomputed from exact rational inputs by the supplementary programs.

Overall assessment of proof architecture

The theorem inputs are settled: the criterion, the effective approximation, and the verified-height input are exactly as the candidate uses them; every analytic argument carries a complete verified proof; and every certified computation has been independently reproduced as described above. The logical architecture is clean: three independent suppliers feed the three hypotheses of a single published criterion, the finite and tail ranges overlap on a full window, the barrier rectangle strictly contains the curved barrier, and every numerical acceptance condition is a strict directed interval inequality that fails closed. The thinnest quantitative margins are the finite-range post-error margin (5.6×10^{-7}), the height-transfer ratio ($1 - \text{ratio} \approx 1.4 \times 10^{-6}$), and the tail contraction ($1 - D \approx 2.8 \times 10^{-4}$); each was reproduced exactly, digit for digit, on the independent toolchain. The remaining trust base is the standard one for computer-assisted proofs of this kind: the correctness of the FLINT/Arb library, of the compiler, and of the line-by-line correspondence between the supplementary programs and the statements of this paper, which the review has checked and which any reader can re-audit from the included sources.

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